

Gaps in Bounded-Shuffle Hierarchies

Counterexamples, arbitrary finite spectra,
and a no-gap theorem for iteration depth

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Abstract

For languages A, B , let $I_k(A, B)$ consist of the words formed by splitting one word of A into at most k pieces and inserting those pieces, in order, into one word of B . The minimum required number of pieces defines an insertion-degree spectrum. We give a negative answer to the proposed assertion that this spectrum must be an initial interval. A ternary example has spectrum $\{1, 3\}$ and satisfies

$$I_1(A, B) = I_2(A, B) = \{a, b, c\}^5 \setminus \{abcba\}, \quad I_3(A, B) = \{a, b, c\}^5.$$

The five-letter exceptional word has the smallest possible length for any gap witness. More generally, we construct finite homogeneous languages realizing every prescribed finite spectrum containing 1, and even prescribe the number of exceptional words added at each degree greater than 1. Thus arbitrarily long plateaus can precede renewed growth. A binary example with spectrum $\{1, 2, 4\}$ shows that two letters already suffice; all of its singleton source-pair spectra are initial intervals. In contrast, the iteration-depth spectrum has no gaps for arbitrary languages, by a shortest-derivation argument. Complete proofs accompany the constructions. Independent exhaustive enumeration and dynamic programming verify the small examples; the accompanying code requires only Python's standard library.

Scope and status. The statements proved below address the indicated no-gap questions, not the surrounding undecidability results. The unrestricted singleton conjecture is not settled here. The proofs have not been checked in a proof assistant or independently peer reviewed. [Added 29 September 2026, batch 41: Part II (Section 11 onwards) realizes every finite spectrum containing 1 over the binary alphabet $\{a, b\}$ with one inserted unary word. It does not settle the singleton conjecture either, and it is not formalized.] [Added 29 September 2026, batch 44: one of those surrounding decision problems is answered in the sibling report `constrained-crossover-closure` of this collection category. Hughes asks for the complexity of deciding $L \otimes_c L = L$ for context-free L , where $\otimes_c$ is constrained (equal-length, equal-cut) one-point crossover [1, Section 4, pp. 3–4]; that report proves the problem co-r.e.-complete over a three-letter alphabet. It concerns crossover, not insertion: it uses nothing proved here, proves nothing about insertion degrees, and leaves the singleton conjecture open. It is unrefereed and not formalized.] [Added 30 September 2026, batch 66: Part III (Section 25 onwards) determines exactly when the single inserted word a^m and a finite binary target language can fill a fixed-letter-count class with exactly one output of degree m and all others of degree 1; the least output lengths for $m = 3, 4, 5$ are 14, 21 and 27. It does not settle the singleton conjecture, and it is not formalized.]

Contents

Part I. Gaps in bounded-shuffle hierarchies	6
1 The problem and the outcome	6
2 Definitions and elementary bounds	6
2.1 Assignment masks	7
2.2 Gaps and plateaus are equivalent	7
3 A five-letter counterexample	8
3.1 All other words have a one-piece representation	8
3.2 The exceptional word requires three pieces	8
4 An arbitrarily delayed final word	9
5 Every finite spectrum can be realized	10
5.1 Choosing positions with a prescribed number of runs	10
5.2 The alphabet and the languages	11
5.3 Proof of the realization theorem	11
5.4 Sizes and an example	12
6 Explicit grammars and automata	12
7 A binary counterexample and representation erasure	13
8 Iteration depth has no gaps	14
8.1 A uniform degree bound along a finite derivation	14
8.2 Why the degree hierarchy behaves differently	15
9 Exact algorithms and reproducibility	15
9.1 Exhaustive assignment enumeration	15
9.2 An independent dynamic program	15
9.3 Verification performed	16
9.4 How the counterexample search was encoded	16
10 What has and has not been settled	17
Part II. Binary insertion spectra from a single unary word	18

11 Provenance, conventions and the answer	18
11.1 One manuscript, one addition	18
11.2 The answer in brief	18
11.3 The repository question and an attribution	19
11.4 Status of the claims	19
11.5 Notation of Part II	20
11.6 Material printed once	20
12 Definitions and main statements	22
12.1 Fixed-letter-count classes	22
13 Gap vectors and the minimum over representations	23
13.1 The exact singleton histogram	24
14 A geometric realization principle	24
15 Explicit binary profiles and exact sizes	26
15.1 Base vectors and separated translates	26
15.2 Exact language cardinalities	27
15.3 All possible finite higher-degree count sequences	28
16 Sharp quadratic size for an isolated high degree	28
16.1 Outputs with no long enough gap	28
16.2 The boundary total cannot attain high degree	28
16.3 An explicit optimal family	29
17 A fully explicit fourteen-symbol gap	30
18 Identical real-rooted singleton histograms, arbitrary aggregate gaps	30
18.1 A Rodrigues polynomial	31
18.2 The aggregate can have almost no real zeros	32
19 Algorithms, succinct descriptions, and certificates	32
19.1 Membership without enumerating the target language	32
19.2 Computing a minimum degree	32
19.3 A literal-word dynamic program	33
20 Executed validation and proof audit	33
20.1 Specific failure modes excluded by the proofs	34
21 Further research questions and topics	34

22 Conclusion of Part II	35
23 A reproducible construction blueprint	36
24 Package contents, rebuilding and the choices of the merge	36
24.1 Shipped files	36
24.2 Rebuilding and rerunning	37
24.3 Where the merge had to choose	37
Part III. Sharp isolation thresholds for binary insertion	38
25 Provenance, conventions and the answer	38
25.1 One manuscript, one addition	38
25.2 The question of Part II	38
25.3 The answer in brief	39
25.4 Why a new criterion is needed	39
25.5 Attribution and limits	40
25.6 Status of the claims	40
25.7 Notation of Part III	40
25.8 Material printed once	40
26 Insertion and finite gap geometry	42
26.1 Conventions	42
27 The largest admissible target language	43
28 The unavoidable volume obstruction	44
29 A complete criterion for an exceptional center	44
29.1 Statement	44
29.2 Reduction to a unique eligible coordinate	45
29.3 Necessity of the local inequalities	45
29.4 Proof of the classification theorem	46
29.5 An effective criterion and negative certificates	46
30 All feasible numerical parameters	47
30.1 A total-mass obstruction beyond the volume bound	47
30.2 Sharp threshold with a prescribed number of gaps	47
31 All extremal centers and their enumeration	49

31.1	Classification at the shortest length	49
31.2	A finite formula for the number of centers	49
32	The boundary family and its sharp cost	50
33	Mandatory targets in every realizing language	51
33.1	The exact covering criterion	51
33.2	A closed test at the sharp length	51
34	Complete classification at degree three	52
34.1	An explicit 37-target realization	54
34.2	What the cardinality result does and does not optimize	54
35	Residue classes and the remaining covering problem	55
36	Reproducible exact checks	56
36.1	What was run	56
36.2	Files and execution	56
37	Proof audit and formalization boundary	57
37.1	Dependencies	57
37.2	Potential failure modes checked explicitly	57
37.3	A realistic formalization route	58
38	Further research questions and topics	58
39	Conclusion of Part III	60
40	Package contents, rebuilding and the choices of the merge	60
40.1	Shipped files	60
40.2	Rebuilding and rerunning	60
40.3	Where the merge had to choose	61
A	The binary finite certificate	61
B	Archive contents and rebuilding	63

1 The problem and the outcome

The question investigated here is whether an increasing hierarchy of bounded interleavings can pause and subsequently grow. It is elementary to state, concerns finite words rather than encodings of computations, and admits exact finite experiments. It also makes a useful distinction between a resource bound and the number of applications of a fixed operation.

Hughes [1, Sections 10–11] formulates an insertion-degree no-gap conjecture, its singleton restriction, a finite representation-erasure problem, and a separate iteration-depth conjecture. The cited version is arXiv:2608.27755v1, submitted 27 August 2026. Searches by title and conjecture terminology on 20 September 2026 found that version and no subsequent resolution; this is a search record, not a guarantee of priority.

Here is the precise answer developed in this note. Among pairs of nonempty finite languages, the possible insertion-degree spectra are *exactly* the finite subsets of the positive integers containing 1. There is no additional interval condition. Moreover, these spectra can be realized with every word of A having one common length and every word of B having another common length. Infinite languages, ambiguous grammars, and unbounded word lengths are unnecessary.

The proof is constructive. It first makes almost every output word easy: a local defect produces a contiguous source factor, hence insertion degree 1. The few outputs with no local defect are rigid, explicitly designated words. Their unique source-position assignments determine their degrees. Selecting those assignments selects the spectrum.

Three complementary constructions make the result transparent. Section 3 gives a small counterexample that can be checked without a computer. Sections 4–5 give the general theorem. Section 7 supplies a binary counterexample and a finite certificate demonstrating that source-pair ambiguity can erase an entire intermediate degree even when each individual pair has no gap. [Added 29 September 2026, batch 41: this paragraph describes Part I. Part II, from Section 11 on, proves a fixed-alphabet version of the realization theorem: over $\{a, b\}$ and with the single inserted word a^m , every finite spectrum containing 1 occurs, with prescribed multiplicities above degree 1 (Theorem 12.1).]

The iteration-depth problem has the opposite answer. Once a word is reached by a shortest sequence of insertions, every intermediate word on that sequence must have the corresponding shortest depth. This statement does not require finiteness, regularity, or an empty-word assumption.

2 Definitions and elementary bounds

Let Σ be a finite alphabet, Σ^* its free monoid, and ε the empty word. The length of a word w is $|w|$. Concatenation is written by juxtaposition. A *factor* is contiguous; a *subsequence* need not be.

Definition 2.1. For $A, B \subseteq \Sigma^*$ and $k \geq 1$, define

$$\mathcal{I}_k(A, B) = \{x_0 y_1 x_1 \cdots y_k x_k : y_1 \cdots y_k \in A, x_0 \cdots x_k \in B, x_i, y_j \in \Sigma^*\}. \quad (1)$$

The inserted source is A and the target source is B . All factors may be empty. The shuffle is

$$A \sqcup B = \bigcup_{k \geq 1} \mathcal{I}_k(A, B).$$

For $w \in A \sqcup B$, put

$$d_{A,B}(w) = \min\{k \geq 1 : w \in \mathcal{I}_k(A, B)\}, \quad \mathcal{D}(A, B) = \{d_{A,B}(w) : w \in A \sqcup B\}.$$

The allowance of empty factors is important. It makes $\mathcal{I}_k(A, B) \subseteq \mathcal{I}_{k+1}(A, B)$, and it ensures that “at most k pieces” and the displayed k -factor definition agree. The degree is oriented: the number counted is the number of runs belonging to A , not the number of runs of both sources combined.

2.1 Assignment masks

For fixed words u, v , a shuffle representation can be recorded by a mask over $\{A, B\}$. Reading the A -positions in order gives u , and reading the B -positions gives v . For instance,

$$\begin{array}{c|ccccc} \text{output} & a & b & c & b & a \\ \text{mask} & A & B & A & B & A \end{array} \quad \text{represents } u = aca, \quad v = bb.$$

A nonempty maximal block of A 's is an A -run.

Lemma 2.2 (Run characterization). *For a nonempty inserted word, the minimum insertion degree of an output is the minimum number of A -runs among all its valid source pairs and assignment masks. An empty inserted word contributes degree 1.*

Proof. A factorization in (1) has at most k nonempty A -runs: empty target pieces may cause two displayed inserted pieces to merge. Conversely, the maximal A -runs of any mask give a factorization of the required form. Minimize in both directions. When the inserted word is empty there are no A -runs, but the definition starts at degree 1. $\square$

For fixed lengths $m = |u|$ and $n = |v|$, there are exactly $\binom{m+n}{m}$ masks. Different masks, and even different source pairs, can produce the same unlabelled output. It is essential to minimize over these representations instead of merely counting the runs in a convenient one. Source-position masks are a particular instance of the trajectory viewpoint on shuffle; the broader framework is developed in [2].

Proposition 2.3 (Elementary restrictions). *The following statements hold.*

- (i) *If A and B are nonempty, then $1 \in \mathcal{D}(A, B)$.*
- (ii) *If $d_{A,B}(w) = r \geq 2$, then $|w| \geq 2r - 1$.*
- (iii) *If $A \subseteq \Sigma^m$, $B \subseteq \Sigma^n$, and $m > 0$, then every degree is at most $\min(m, n + 1)$.*
- (iv) *Over a one-letter alphabet, nonempty language pairs have spectrum $\{1\}$.*

Proof. For (i), choose $u \in A, v \in B$; the concatenation uv has a one-piece representation. For (ii), a minimal mask with r A -runs contains at least r A -positions and at least $r - 1$ separating B -positions. For (iii), there are at most m nonempty A -runs and at most $n + 1$ gaps around the target letters. For (iv), every shuffle of a^i and a^j is the concatenation a^{i+j} and therefore has degree 1. $\square$

2.2 Gaps and plateaus are equivalent

The spectrum records exactly when new output words appear:

$$k \in \mathcal{D}(A, B) \iff \mathcal{I}_{k-1}(A, B) \subsetneq \mathcal{I}_k(A, B) \quad (k \geq 2). \quad (2)$$

Thus $\mathcal{I}_k(A, B) = \mathcal{I}_{k+1}(A, B)$ says only that degree $k + 1$ is absent. It says nothing purely set-theoretic about later degrees. The no-gap conjecture supplies precisely the missing assertion; the constructions below disprove it.

3 A five-letter counterexample

Let $\Sigma = \{a, b, c\}$, and set

$$F = \{aba, abb, abc, acb, bba, bca, bcb, cba\}, \quad A = \Sigma^3 \setminus F, \quad B = \Sigma^2. \quad (3)$$

Thus $|A| = 19$ and $|B| = 9$. Explicitly,

$$A = \{aaa, aab, aac, aca, acc, baa, bab, bac, bbb, bbc, bcc, caa, cab, cac, cbb, cbc, cca, ccb, ccc\}.$$

Theorem 3.1. *For the languages in (3),*

$$\begin{aligned} \mathcal{I}_1(A, B) &= \mathcal{I}_2(A, B) = \Sigma^5 \setminus \{abcba\}, \\ \mathcal{I}_k(A, B) &= \Sigma^5 \quad (k \geq 3). \end{aligned}$$

Consequently $\mathcal{D}(A, B) = \{1, 3\}$.

3.1 All other words have a one-piece representation

A five-letter word belongs to $\mathcal{I}_1(A, B)$ exactly when one of its three length-three factors belongs to A . Indeed, remove that contiguous factor; the remaining two letters form a word of B . Conversely, a one-piece insertion exhibits such a factor.

We therefore need to find the five-letter words all of whose length-three factors belong to F . Regard a triple $xyz \in F$ as a directed edge from xy to yz . The complete edge table is

Vertex	Outgoing vertices
ab	ba, bb, bc
ac	cb
bb	ba
bc	ca, cb
cb	ba
ba, ca	none

Other two-letter vertices have no outgoing edges. The only path consisting of three edges is

$$ab \longrightarrow bc \longrightarrow cb \longrightarrow ba.$$

It spells $abcba$. Thus every other five-letter word has an A -factor and has degree 1. This proves the first equality in

$$\mathcal{I}_1(A, B) = \Sigma^5 \setminus \{abcba\} \subseteq \mathcal{I}_2(A, B) \subseteq \mathcal{I}_3(A, B) \subseteq \Sigma^5.$$

3.2 The exceptional word requires three pieces

The distinct length-three subsequences of $abcba$ are

$$abc, abb, aba, acb, aca, bcb, bca, bba, cba.$$

Exactly one is in A : aca . Its occurrence is unique, at positions 1, 3, 5. The complementary source is bb , so its mask is $ABABA$ and has three A -runs. There is no alternative source pair or alignment that could lower this cost. Hence

$$d_{A,B}(abcba) = 3.$$

All output words have length five, so this completes the proof of Theorem 3.1.

Corollary 3.2 (Minimum possible witness length). *Five is the smallest length of a word that can witness a gap in an insertion-degree spectrum.*

Proof. Degree 1 cannot be missing for a nonempty pair. Any gap must therefore have a word of degree at least 3 above it, whose length is at least 5 by Proposition 2.3. The example attains this bound. $\square$

This does not assert that 19 and 9 are minimum language sizes. The exact optimality claim concerns witness length only.

4 An arbitrarily delayed final word

The small example suggests a more systematic device: make all but one full-length word contain an easy contiguous source factor.

Fix $r \geq 2$, put $N = 2r - 1$, and use the ordered alphabet

$$\Sigma_r = \{1, 2, \dots, N\}.$$

These integers denote individual alphabet symbols, not decimal strings. Let

$$W_r = 1\,2\,\dots\,N, \quad U_r = 1\,3\,5\,\dots\,N.$$

Define

$$\begin{aligned} A_r &= \{u \in \Sigma_r^r : u \text{ is not strictly increasing}\} \cup \{U_r\}, \\ B_r &= \Sigma_r^{r-1}. \end{aligned} \tag{4}$$

Lemma 4.1 (A local defect supplies a factor). *Every word $w \in \Sigma_r^N$ other than W_r has a length- r factor that is not strictly increasing.*

Proof. A strictly increasing word of length N over an N -letter ordered alphabet is necessarily W_r . Thus $w \neq W_r$ has an adjacent pair $w_i \geq w_{i+1}$. Any adjacent pair in a word of length at least $r \geq 2$ is contained in a factor of length r : extend the pair to the left and right within the word until the length is r . That factor is not strictly increasing. $\square$

Theorem 4.2 (One-word delay). *For every $r \geq 2$, the pair in (4) satisfies*

$$\mathcal{I}_k(A_r, B_r) = \begin{cases} \Sigma_r^N \setminus \{W_r\}, & 1 \leq k < r, \\ \Sigma_r^N, & k \geq r. \end{cases} \tag{5}$$

In particular, $\mathcal{D}(A_r, B_r) = \{1, r\}$.

Proof. For $w \neq W_r$, Lemma 4.1 provides a contiguous factor in A_r . Deleting it leaves $r - 1$ symbols, automatically a word of B_r . Hence $d_{A_r, B_r}(w) = 1$.

Every subsequence of W_r is strictly increasing. The only length- r subsequence accepted by A_r is therefore U_r . Since all letters of W_r are distinct, this subsequence has exactly one alignment, using positions $1, 3, \dots, 2r - 1$. Its complement is in B_r , and its mask has exactly r A -runs. Thus $d_{A_r, B_r}(W_r) = r$. All shuffles have length N , and the displayed formula follows. $\square$

The language sizes and stage sizes are explicit:

$$\begin{aligned} |A_r| &= (2r - 1)^r - \binom{2r - 1}{r} + 1, \\ |B_r| &= (2r - 1)^{r-1}, \\ |\mathcal{I}_k(A_r, B_r)| &= (2r - 1)^{2r-1} - 1 \quad (1 \leq k < r). \end{aligned}$$

For $r = 3$, this gives an especially simple ordered-alphabet construction with $|A_3| = 116$, $|B_3| = 25$, and a single exceptional word among 3125 possible outputs. Section 3 uses fewer symbols and fewer source words, but the ordered construction scales immediately.

Corollary 4.3 (No fixed plateau test is sound). *There is no constant $h \geq 1$ such that, for all finite language pairs, observing h consecutive equalities in the degree hierarchy guarantees permanent stabilization.*

Proof. Take $r = h + 2$. Then stages 1 through $h + 1$ are equal, giving h consecutive equalities, but stage $h + 2$ adds W_r . $\square$

This is a statement about tests based solely on a fixed number of repeated stages, not an undecidability claim. For explicitly finite input languages, the elementary length bound still supplies a termination bound.

5 Every finite spectrum can be realized

We now prescribe several exceptional outputs independently. Disjoint alphabet blocks prevent one exception from supplying a representation of another.

Theorem 5.1 (Finite delay-profile realization). *Let $t_1, \dots, t_q$ be any nonempty finite list of integers at least 2, and choose $m \geq \max_i t_i$. There are a finite alphabet Σ , finite languages*

$$A \subseteq \Sigma^m, \quad B = \Sigma^{m-1},$$

and distinct words $W_1, \dots, W_q \in \Sigma^{2m-1}$ such that

$$\mathcal{I}_k(A, B) = \Sigma^{2m-1} \setminus \{W_i : t_i > k\} \quad (k \geq 1). \quad (6)$$

Thus $\mathcal{D}(A, B) = \{1, t_1, \dots, t_q\}$, and the number of words first added at degree $k \geq 2$ is exactly $|\{i : t_i = k\}|$.

5.1 Choosing positions with a prescribed number of runs

Set $N = 2m - 1$. For $1 \leq t \leq m$, define a subset of positions

$$J_{m,t} = \{1, \dots, m - t + 1\} \cup \{m - t + 1 + 2j : 1 \leq j \leq t - 1\}. \quad (7)$$

It has $(m - t + 1) + (t - 1) = m$ elements. Its largest position is $m + t - 1 \leq N$. Its first $m - t + 1$ elements form one interval, and each of its remaining $t - 1$ elements is separated from the preceding selected position by one unselected position. Hence its indicator mask has exactly t runs.

For $m = 4$, three examples are

t	$J_{4,t}$	mask
2	$\{1, 2, 3, 5\}$	AAABABB
3	$\{1, 2, 4, 6\}$	AABABAB
4	$\{1, 3, 5, 7\}$	ABABABA

This explicit position formula avoids any existential choice in the construction.

5.2 The alphabet and the languages

Use q disjoint ordered blocks

$$C_i = \{c_{i,1}, \dots, c_{i,N}\}, \quad \Sigma = C_1 \dot{\cup} \dots \dot{\cup} C_q,$$

and define

$$W_i = c_{i,1}c_{i,2} \cdots c_{i,N}.$$

Call a word *block-increasing* if all its letters belong to one block C_i and their second indices are strictly increasing. Let U_i be the subsequence of W_i at the positions J_{m,t_i} . Finally set

$$A = \{u \in \Sigma^m : u \text{ is not block-increasing}\} \cup \{U_1, \dots, U_q\}, \quad B = \Sigma^{m-1}. \quad (8)$$

5.3 Proof of the realization theorem

Suppose $w \in \Sigma^N$ is not one of the W_i . If every adjacent pair of w belonged to a common block and increased within that block, all letters of w would lie in one block and form a strictly increasing sequence. Since its length is the block size N , it would be that block's W_i , a contradiction.

Therefore w contains an adjacent pair that either crosses between blocks or fails to increase. Extend that pair to a length- m factor. This factor is not block-increasing, belongs to A , and leaves a complementary word of length $m - 1$ in B . Consequently,

$$w \notin \{W_1, \dots, W_q\} \implies d_{A,B}(w) = 1.$$

Now fix W_i . Every one of its subsequences is block-increasing within C_i . The only such length- m word accepted by A is U_i : the other exceptional words U_j use disjoint alphabets. The letters of W_i are distinct, so the alignment of U_i is unique. Its selected positions are exactly J_{m,t_i} and have t_i runs. The complementary word has length $m - 1$, so it belongs to B . Hence

$$d_{A,B}(W_i) = t_i.$$

These two calculations prove (6). □

Corollary 5.2 (Complete classification for finite languages). *A set $S \subseteq \mathbb{N}$ is the insertion-degree spectrum of a pair of nonempty finite languages over some finite alphabet if and only if S is finite and $1 \in S$.*

Proof. Necessity follows because a shuffle of two finite languages is finite and because degree 1 is always attained. If $S = \{1\}$, take $A = \{a\}$ and $B = \{\varepsilon\}$. Otherwise list the elements of $S \setminus \{1\}$ as $t_1, \dots, t_q$ and apply Theorem 5.1 with $m = \max S$. $\square$

The empty spectrum is realized by an empty source language. Thus, allowing empty inputs, the possible finite spectra are exactly $\emptyset$ and the finite sets containing 1.

5.4 Sizes and an example

The construction has qN alphabet symbols. Exactly $q \binom{N}{m}$ length- m words are block-increasing; we add back q selected words. Therefore

$$|A| = (qN)^m - q \binom{N}{m} + q, \quad |B| = (qN)^{m-1},$$

$$|\mathcal{I}_k(A, B)| = (qN)^N - |\{i : t_i > k\}|.$$

For example, choose the desired spectrum $\{1, 2, 4\}$, use two seven-letter blocks, and set $m = 4$. From the first block select positions 1, 2, 3, 5; from the second select 1, 3, 5, 7. All length-seven outputs except the two increasing block words have degree 1. The first exceptional word appears at degree 2, degree 3 adds nothing, and the second appears at degree 4.

Repeated values in the delay list are also meaningful. The list $(2, 3, 3, 4)$ adds one exceptional word at degree 2, two at degree 3, and one at degree 4. Thus the theorem controls the positive-degree increments, not merely their support.

6 Explicit grammars and automata

Although the displayed finite sets may be large, their descriptions need not enumerate every accepted word. This section gives direct language descriptions for the realization theorem.

Let E be the set of *bad adjacent pairs*: ordered symbol pairs that either come from different blocks or fail to increase within a common block. The non-block-increasing part of A is exactly

$$A_{\text{bad}} = \bigcup_{j=0}^{m-2} \Sigma^j E \Sigma^{m-j-2}. \quad (9)$$

Thus $A = A_{\text{bad}} \cup \{U_1, \dots, U_q\}$ is given by a finite regular expression, and $B = \Sigma^{m-1}$ is even simpler. The equivalence follows because a word is block-increasing exactly when every adjacent pair is good.

For a grammar description, use a nonterminal X with productions $X \rightarrow c$ for $c \in \Sigma$. Let $P_0 \rightarrow \varepsilon$ and $P_j \rightarrow XP_{j-1}$ for $1 \leq j \leq m-1$. A nonterminal E_0 has productions $E_0 \rightarrow cd$ for bad pairs $(c, d) \in E$. Then take

$$\begin{aligned} S_A &\rightarrow P_j E_0 P_{m-j-2} & (0 \leq j \leq m-2), \\ S_A &\rightarrow U_i & (1 \leq i \leq q), \\ S_B &\rightarrow P_{m-1}. \end{aligned}$$

This is a finite acyclic context-free grammar for the required finite languages. It may have several parses when a word contains several bad pairs; that ambiguity is irrelevant to the insertion-degree definition. Regularity follows directly from (9), independently of the grammar presentation.

Writing $Q = |\Sigma|$, the grammar uses $O(m + Q^2 + qm)$ total right-hand-side symbols when the exceptional words are written explicitly. A direct automaton for A_{bad} tracks the length read, the preceding alphabet symbol, and whether a bad pair has occurred. Accepting at length m in the bad state recognizes A_{bad} . A finite trie for the U_i supplies the exceptional part. These are construction bounds, not claims of minimal grammar or automaton size.

For each fixed k , one may also recognize $\mathcal{I}_k(A, B)$ by a finite automaton that nondeterministically assigns each input symbol to an A -automaton or a B -automaton, remembers the previous source, and counts starts of A -runs up to k . This automaton is finite whenever the source automata are finite. The gap phenomenon is therefore not caused by stepping outside regular languages.

7 A binary counterexample and representation erasure

The parametric theorem uses an alphabet that grows with the prescribed profile. A separate small example demonstrates that a binary alphabet already supports a gap. [Added 29 September 2026, batch 41: Part II removes the growing alphabet altogether: every finite spectrum containing 1 is realized over $\{a, b\}$ (Corollary 12.2).] Set

$$A = \{aaba, aabb, abbb, bbaa, bbba, bbbb\}, \quad B = \{aaa, aba, bab\}. \quad (10)$$

All shuffles have length seven, and each source pair has $\binom{7}{4} = 35$ assignment masks.

Theorem 7.1. *For (10), the degree counts are*

k	1	2	3	4
$ \{w : d_{A,B}(w) = k\} $	57	53	0	1.

The unique degree-four word is bababab. Every singleton pair $(u, v) \in A \times B$ has an initial-interval spectrum.

A finite certificate is given in Appendix A, together with complete machine-readable degree and source-pair tables. The proof is an exhaustive check of $18 \cdot 35 = 630$ masks followed by minimization over equal outputs. It has also been checked independently by the dynamic program in Section 9. The analytical realization theorem does not depend on this finite computation.

There is a particularly short certificate for the absence of degree 3. Across the eighteen singleton source pairs, exactly twenty-three distinct words have singleton minimum degree 3. Appendix A gives a source pair and a mask with at most two A -runs for each of those words. Therefore every candidate language-level degree-three word loses that degree to a competing representation. The remaining degree-four witness does not.

For *bababab*, a source-pair check leaves only $(bbbb, aaa)$; its unique mask is *ABABABA*. To see the exclusion explicitly, a source *aba* would leave either *abbb* or *bbba* by letter counts. The only possible projections of those forms leave respectively *baa* and *aab*, not *aba*. A source *bab* would leave either *aabb* or *bbaa*; their possible projections leave respectively *bba* and *abb*, not *bab*. The other source combinations fail letter counts. Thus the high-degree witness survives.

Corollary 7.2. *Competing source-pair representations can create a gap even when every singleton source-pair spectrum is an initial interval. Moreover, two is the minimum alphabet size admitting any insertion-degree gap.*

Proof. The first assertion is exactly the binary example and its pair table, answering the finite representation-erasure question affirmatively. The example proves the binary upper bound; the unary impossibility is Proposition 2.3(iv). $\square$

This does not disprove the singleton no-gap conjecture: the source languages in the counterexample have six and three words. It disproves the inference from good behavior of every individual source pair to good behavior of their union. [Added 29 September 2026, batch 41: Part II's constructions reduce the inserted language to the single word a^m , but still use many target words, so they do not disprove it either (Question 21.1).]

8 Iteration depth has no gaps

Now hold the operation fixed and count applications rather than pieces. Put

$$X_0 = B, \quad X_{m+1} = A \sqcup X_m,$$

and, for a word reachable at some finite stage, define

$$\iota_{A,B}(w) = \min\{m \geq 0 : w \in X_m\}, \quad \mathcal{T}(A, B) = \{\iota_{A,B}(w) : w \in \bigcup_{m \geq 0} X_m\}.$$

No monotonicity of the exact stages X_m is assumed. For example, $A = \{a\}$ and $B = \{\varepsilon\}$ give $X_m = \{a^m\}$.

Lemma 8.1 (Shortest-prefix lemma). *Let R be any binary relation on a set V , and let $B \subseteq V$. If a vertex v has shortest R -path length r from B , there is a vertex of shortest path length j for every $0 \leq j \leq r$.*

Proof. Choose a shortest path $v_0 R v_1 R \cdots R v_r = v$ with $v_0 \in B$. Its prefix reaches v_j in j steps. A path to v_j of length $\ell < j$, followed by the suffix $v_j R \cdots R v_r$, would reach v in $\ell + r - j < r$ steps. This contradicts the choice of the path. Thus v_j has shortest length j . $\square$

Theorem 8.2 (Iteration-depth no-gap theorem). *For arbitrary languages A, B , $\mathcal{T}(A, B)$ is an initial interval of $\mathbb{N}_0$, possibly empty or infinite. Equivalently,*

$$r \in \mathcal{T}(A, B) \implies \{0, 1, \dots, r\} \subseteq \mathcal{T}(A, B).$$

The same statement holds when each insertion is restricted to any one fixed degree bound.

Proof. Take $V = \Sigma^*$ and let xRy mean $y \in A \sqcup \{x\}$. The least number of relation steps from B is exactly $\iota_{A,B}$, so apply Lemma 8.1. For a fixed degree bound k , replace the relation by $y \in \mathcal{I}_k(A, \{x\})$. $\square$

8.1 A uniform degree bound along a finite derivation

One formulation of the depth parameter minimizes m subject to the existence of a single finite k usable in every step of an m -step derivation. This gives the same depth as unrestricted shuffle at each step. One direction is immediate. For the other, a finite derivation uses finitely many assignment masks, hence finitely many degrees $k_1, \dots, k_m$. Their maximum K bounds every step, since empty pieces pad a representation to degree K . The case $m = 0$ uses any $K \geq 1$.

Thus Theorem 8.2 applies to that formulation as well. Its argument does not require a largest word in either language. It also handles $B = \emptyset$, when the spectrum is empty, and $A = \emptyset$ with $B \neq \emptyset$, when the spectrum is $\{0\}$.

8.2 Why the degree hierarchy behaves differently

For an autonomous iteration $Y_{m+1} = F(Y_m)$, an equality $Y_m = Y_{m+1}$ immediately gives equality at every later stage, simply by applying F . If cumulative reachability is wanted, use

$$C_m = \bigcup_{j=0}^m X_j, \quad C_{m+1} = C_m \cup (A \sqcup C_m).$$

Shuffle distributes over unions in its second argument, so this is another autonomous iteration.

By contrast, $I_k(A, B)$ increases the admissible number of source runs in a *single* shuffle. It is not obtained by applying one fixed extension operator to the previous unlabelled output set. Minimizing over representations may discard the source information needed to extend a particular witness. The counterexamples show that this loss of information can hide every witness at one cost while preserving a witness at a later cost.

9 Exact algorithms and reproducibility

9.1 Exhaustive assignment enumeration

For each $u \in A$, $v \in B$, enumerate every $|u|$ -element subset of the $|u| + |v|$ output positions. Merge the words according to that mask, count its A -runs, and retain the smallest count for each output word. A second minimum over source pairs gives the language-level degree.

With $m = |u|$ and $n = |v|$, this takes

$$O\left((m+n) \binom{m+n}{m}\right)$$

character operations per source pair, aside from ordinary dictionary costs. The implementation normalizes the empty-inserted-word case to degree 1 and treats an absent output as nonmembership, not as degree zero.

The raw masks are witnesses only of upper bounds. Exactness comes from exhaustive coverage of every mask and every source pair. This distinction is particularly important in a problem whose obstruction is competing representations.

9.2 An independent dynamic program

For a fixed triple (u, v, w) with $|u| = m$, $|v| = n$, and $|w| = m + n$, define $D(i, j, s)$ as the minimum number of A -runs after matching the first i letters of u and j letters of v against the prefix of w of length $i + j$. The value $s \in \{A, B\}$ records the most recently used source; the starting state is treated as B .

Initialize $D(0, 0, B) = 0$ and all other states to $+\infty$. If $u_{i+1} = w_{i+j+1}$, relax

$$D(i+1, j, A) \leftarrow \min(D(i+1, j, A), D(i, j, s) + [s \neq A]).$$

If $v_{j+1} = w_{i+j+1}$, relax

$$D(i, j+1, B) \leftarrow \min(D(i, j+1, B), D(i, j, s)).$$

At the end, take $\min_s D(m, n, s)$ and normalize zero to one. Infinite cost means that w is not a shuffle of the fixed pair.

Proposition 9.1. *The dynamic program computes the exact singleton insertion degree in $O((m+1)(n+1))$ time and space.*

Proof. Every transition consumes exactly the next letter of one source and the next letter of the output. Paths therefore correspond bijectively to valid assignment masks. A transition into source A charges one exactly at the start of an A -run. The state graph is acyclic because $i + j$ increases at every transition. Induction in that sum proves the recurrence computes minimum path cost. There are $2(m + 1)(n + 1)$ states and at most two outgoing transitions per state. $\square$

Minimizing the dynamic-programming result over all pairs gives a second algorithm for $d_{A,B}(w)$. The verifier compares these two implementations for *every* word in the complete small output universes, including words outside the shuffle.

9.3 Verification performed

The following counts come from the included run of `code/verify.py`.

Example	Source pairs	Masks	DP queries	Shuffle words
Ternary, lengths (3, 2)	171	1710	41553	243
Binary, lengths (4, 3)	18	630	2304	111

The ternary singleton spectra comprise 27 copies of $\{1\}$, 106 copies of $\{1, 2\}$, and 38 copies of $\{1, 2, 3\}$. The archive also contains every individual pair’s spectrum and degree histogram.

For the general ordered construction, all 27 output words at $r = 2$ and all 3125 output words at $r = 3$ were checked by enumerating every possible source-position set. Larger single-block and multiblock instances were checked on all target words, all their single-letter substitutions, all adjacent swaps of targets, and reproducibly seeded random samples. Those larger checks are explicitly samples, not exhaustive tests or substitutes for the theorem’s proof. Their exact counts are recorded in `data/verification.json`.

An additional finite check enumerates all 52 equality patterns among the five source positions of a pair of lengths (3, 2). It confirms the singleton no-gap property for every such pair over any alphabet: relabelling distinct symbols does not change the mask equalities. This bounded-length check should not be confused with a proof of the unrestricted singleton conjecture.

9.4 How the counterexample search was encoded

The initial search used Boolean variables a_u, b_v selecting words into A and B . For each potential output w , precompute its exact singleton degree for every potential pair. Define

$$Q_{w,k} = \bigvee_{d_{u,v}(w) \leq k} (a_u \wedge b_v).$$

To forbid degree g , impose $Q_{w,g} \Rightarrow Q_{w,g-1}$ for every output. To require a larger degree, require at least one w with $Q_{w,M} \wedge \neg Q_{w,g}$, where M is the length-based maximum degree. These constraints express the actual minima, not merely a chosen mask’s run count.

A Z3 search suggested the small examples; the general construction was subsequently proved without relying on the solver. The archive includes a portable SMT-LIB instance and its generator. Neither Z3 nor the discovery search is needed to run the exact verifier. No minimality claim about the total number of source words rests on the solver’s output.

10 What has and has not been settled

The insertion-degree no-gap assertion is false, already for finite regular languages. The finite representation-erasure question has an affirmative answer: the binary certificate exhibits the required source-pair behavior. The general finite-spectrum realization theorem strengthens the negative answer from an isolated gap to an arbitrary finite profile. Separately, the iteration-depth no-gap assertion is true for arbitrary languages.

The unrestricted singleton insertion-degree question remains outside the proofs in this note. The arbitrary-spectrum construction intentionally exploits multiple possible source words and cannot be read as a singleton counterexample. Similarly, the argument does not classify infinite spectra over a fixed finite alphabet, and it does not establish which finite spectra can all be realized over one fixed binary alphabet. [Added 29 September 2026, batch 41: the last point is settled in Part II. Every finite set of positive integers containing 1 is the spectrum of a pair of finite binary languages, with the inserted language the single unary word $\{a^m\}$ (Theorem 12.1, Corollary 12.2). The unrestricted singleton question and the classification of infinite spectra over a fixed finite alphabet remain open (Questions 21.1 and 21.9).]

Several concrete problems are left by these results. One can seek smaller source languages for a specified spectrum, determine the least alphabet needed to realize $\{1, r\}$ as r grows, or study descriptions by small deterministic automata rather than by cardinality. The proof of Theorem 5.1 also suggests investigating classes where competing source-pair representations are controlled strongly enough to restore an interval theorem. These are research directions, not additional claims of unresolved status in the published literature. [Added 29 September 2026, batch 41: Part II answers the least-alphabet question: for every $r \geq 2$ the spectrum $\{1, r\}$ is realized over two letters, and no fewer suffice (Corollary 12.2). It answers the smallest-source question for the inserted language only, which it reduces to one word; the minimum size of the target language stays open (Question 21.2), as do small automaton descriptions (Question 21.7) and interval-restoring classes (Question 21.8). Its quadratic length bound (Theorem 12.3) holds only in its full-coverage class, not for every binary spectrum.]

The central lesson is specific: an insertion-degree spectrum is a minimum over representations, and taking that minimum can erase intermediate costs. The same phenomenon does not affect shortest derivation depth, because a cheaper prefix could always be substituted into a derivation. This difference accounts for the opposite answers without appealing to general undecidability machinery.

Part II. Binary insertion spectra from a single unary word

Sections 1–10 and Appendices A and B form the original report (Part I) of 20 September 2026, unchanged except for dated pointers to this addition. Part II was added on 29 September 2026 in batch 41 of ProveIt’s incoming-report intake, from one manuscript written against the state of Part I printed above. It answers the fixed-binary question that Section 10 left open: every finite set of positive integers containing 1 is the insertion-degree spectrum of a pair of finite binary languages, even with the inserted language a single unary word $\{a^m\}$. Section 11 records the provenance and fixes the notation of the addition.

11 Provenance, conventions and the answer

11.1 One manuscript, one addition

Part II is batch-41 manuscript 01, *Binary Insertion Spectra from a Single Unary Word: Arbitrary finite profiles and a sharp quadratic witness bound*, a research draft prepared with ChatGPT for Vladimir Reshetnikov and dated 28 September 2026 (18 pages as delivered). It arrived as the archive `ProveIt_Binary_Insertion_Spectra.zip` (inner directory of the same name) in ProveIt commit `9754e8360` and was placed in `eaf787d50`. It was written against ProveIt commit `e2b1f016a` (in full `e2b1f016a94102663f12b970e6dd434229f90d01`), at which Part I’s `article.tex` had the Git blob `946173d8`; that blob is unchanged at the placement commit, so every statement the manuscript makes about Part I refers to the text printed above. Its code, data and audit files are shipped beside Part I’s files with the prefix `02-binary-profiles-` (Part I’s unprefix files count as 01); its manuscript, delivery README and PDF are not shipped and survive in the arrival commit.

The manuscript contributes, in the order printed here: the main statements, namely arbitrary finite profiles over $\{a, b\}$ with one unary inserted word, the fixed-binary classification, and a sharp length bound in the full-coverage class (Section 12); the gap-vector formula for the degree and the exact singleton histogram (Section 13); a geometric realization principle (Section 14); the explicit construction with exact sizes and arbitrary higher-degree multiplicities (Section 15); the sharp quadratic lower bound and its optimal family (Section 16); a fourteen-symbol worked example (Section 17); real-rooted singleton histograms against arbitrary aggregate gaps (Section 18); algorithms (Section 19); its executed checks (Section 20); ten research questions (Section 21); its conclusion (Section 22); a construction blueprint (Section 23); and its package notes (Section 24). Every result, proof, example, remark, limitation and question of the manuscript is printed. The definitions it repeats from Part I are printed once, in Part I, and cited (Section 11.6). Remark 11.1 and the re-scoped passages of Part I were added in the merge.

11.2 The answer in brief

Part I realizes arbitrary finite insertion-degree spectra over alphabets that grow with the requested profile (Theorem 5.1) and leaves the fixed binary alphabet open (Section 10). Part II proves that every finite subset of the positive integers containing 1 is realizable over $\{a, b\}$, even with the inserted language equal to the single word $\{a^m\}$. More strongly, any prescribed finite multiplicities of degrees $2, \dots, m$ can be realized, with every other output having degree 1. Both inputs are finite and homogeneous, and their shuffle is an entire fixed-letter-count class.

The proof turns a binary word into its vector of a -gap lengths. Insertion degree becomes the minimum support size of a nonnegative vector difference. Separated exceptional vectors protect the desired degrees, while a quantitative escape lemma makes every other output a one-piece insertion. This mechanism requires no search oracle or unproved singleton conjecture.

Part II also proves an exact extremal result. With $A = \{a^m\}$, if the shuffle is an entire binary fixed-letter-count class, every output but one has degree 1, and the remaining output has degree $m \geq 3$, then its length is at least m^2 . For every $m \geq 6$, the construction attains m^2 . *The full-coverage hypothesis is essential to the scope of this bound: it is not a lower bound on the length of an arbitrary binary word of insertion degree m , for which the elementary bound is $2m - 1$ (Proposition 2.3(ii)).*

Each individual source pair nevertheless has the same gap-free degree histogram. Part II computes that histogram and proves its real-rootedness in the balanced case, while the combined language can have arbitrary missing degrees and almost entirely nonreal histogram zeros. Full proofs, an explicit 14-symbol example, exact verification code, executed receipts, and further research questions are included.

11.3 The repository question and an attribution

The manuscript read Part I at the pin as follows. Part I proves arbitrary finite-spectrum realization with an alphabet that grows with the requested profile (Theorem 5.1), and supplies a binary example with spectrum $\{1, 2, 4\}$ (Theorem 7.1). It explicitly leaves open the classification over a fixed binary alphabet and the least alphabet size for spectra $\{1, r\}$ (Section 10). The earlier realization theorem is not claimed as new here.

Part II proves a stronger fixed-alphabet statement. The inserted language can contain just one word, consisting entirely of copies of one letter; all the source-pair ambiguity can be confined to the target language. The target words have the same length and the same number of each letter. Moreover, one can prescribe not only which degrees occur but the exact number of words of each degree above 1.

Hughes formulates the general insertion-degree no-gap conjecture, its both-singleton restriction, and a finite representation-erasure problem in Section 11 of [1]. The same source already proves that a unary inserted word and a single arbitrary target word have an initial-interval spectrum [1, Theorem 9, page 9]. Part II uses and reproves its gap-distribution observation; *the unary-source singleton interval theorem is not a new contribution of this addition.*

The new result is that an arbitrary finite spectrum can arise by taking the minimum over a finite family of these particularly well-behaved singleton pairs, while keeping the inserted word fixed. It is a uniform strengthening of the representation-erasure phenomenon already exhibited in Part I (Section 7). It is not a resolution of the unrestricted both-singleton conjecture.

11.4 Status of the claims

The manuscript inspected the repository at commit e2b1f016a, in particular Part I's source

```
SetTheory/Cardinals/docs/reports/
automata-and-formal-languages/insertion-degree-spectra/article.tex
```

(its definitions, finite-profile construction, binary example, algorithm and validation discussion, and closing limitations), together with Part I's README and the collection's report manifest. The primary preprint consulted was arXiv:2608.27755v1; its definitions and Section 11 were checked directly, including the rendered ninth page. A targeted repository search and web searches did not identify the fixed-binary, single-unary-source profile theorem or the extremal theorem below. This is a bounded search record, not

a guarantee of priority. The technical arguments are supplied in full so their validity does not depend on novelty assertions or on the correctness of unrelated repository drafts. All constructions and checks of the manuscript are independent of a repository checkout.

Status	Content
Established background	Unary-source singleton no-gap theorem [1, Theorem 9]; Part I’s variable-alphabet realization and binary gap example.
Proved in Part II	Fixed-binary realization of arbitrary finite profiles with $A = \{a^m\}$; explicit size formulas; sharp m^2 isolated-witness length for $m \geq 6$ in the full-coverage class.
Supporting developments	A gap-vector minimization formula, a separated-center construction, exact singleton histograms, and a self-contained real-rootedness proof.
Not established	The unrestricted both-singleton conjecture; optimal target-language cardinality; general infinite spectra; exact optimal lengths for $m = 3, 4, 5$; proof-assistant verification or independent peer review.

[Added 30 September 2026, batch 66: two entries of the last row are settled or narrowed in Part III. The exact optimal lengths for $m = 3, 4, 5$ are 14, 21 and 27 (Theorem 30.3). The optimal target-language cardinality is determined only for $m = 3$ at the shortest length, where it is 37 (Theorem 34.1). Part III is unrefereed and not formalized either.]

Status (manuscript 01). The new theorems are unrefereed research claims with written proofs, not Lean-verified results. Nothing in Part II, and nothing in Part I, is formalized in any proof assistant; the report’s placement in a collection that sits beside Lean and Rocq developments confers no formal status on it. The finite computations of Section 20 are additional exact checks, not inductive evidence substituted for the proofs.

11.5 Notation of Part II

Part I’s symbols keep their meanings. In particular $\mathcal{I}_k(A, B)$, $A \sqcup B$, $d_{A,B}(w)$ and $\mathcal{D}(A, B)$ are those of Section 2; the manuscript’s definitions of them agree with Part I’s, orientation included (A is the inserted source, B the target source), and follow [1, Sections 2 and 11]. Where a symbol of the manuscript clashes with a symbol of Part I, Part II renames it; Table 1 lists the new symbols, the renamings, and the tempting false readings. No normalization was changed. Equations of Part II continue Part I’s single equation sequence, and theorem-like items are numbered by section, so the numbers differ from those of the delivered manuscript.

11.6 Material printed once

The manuscript restates Part I’s definitions: the k -insertion $\mathcal{I}_k(A, B)$ with empty factors allowed (1), the shuffle, the degree $d_{A,B}$ and the spectrum $\mathcal{D}(A, B)$, and the characterization of the degree as the minimum number of inserted-source runs over all source pairs and all valid masks. These are printed once, in Section 2; the run characterization is Lemma 2.2, which the manuscript proves by the same argument (maximal runs give one direction, the nonempty inserted factors of (1) the other). The manuscript’s degree-one argument for a one-letter alphabet is Proposition 2.3(iv), and its remark that allowing empty input languages adds precisely the empty spectrum is Part I’s remark after Corollary 5.2. Its literal-word dynamic program (Section 19.3) is Part I’s dynamic program of Section 9 in another encoding; it is printed as the manuscript gives it, with that identification noted.

Remark 11.1 (Two realizations compared; added in the merge). Part I’s Theorem 5.1 realizes a profile with largest degree m at output length $2m - 1$, the least length any word of degree m can have (Proposition 2.3(ii)), over an alphabet of $q(2m - 1)$ symbols; its shuffle is all of Σ^{2m-1} . Part II fixes the alphabet $\{a, b\}$ and

Table 1: Symbols of Part II. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

Symbol	Meaning in Part II	Not to be confused with
$A = \{a^m\}, B$	the inserted source, one unary word, and the target language (same roles as in Section 2)	Part I’s particular languages (3), (4), (8), (10)
m, q, t_i, W_i	inserted length; number of prescribed degrees; the list $t_0, \dots, t_{q-1}$; the exceptional outputs	the same roles in Theorem 5.1, whose list is $t_1, \dots, t_q$ and whose W_i lie in Σ^{2m-1} over a growing alphabet
g (renamed from d)	number of gaps of a binary word, one more than its number of b ’s	the degree $d_{A,B}(w)$
T	number of a ’s in an output	$\mathcal{T}(A, B)$, the iteration-depth spectrum of Section 8
$N = T + g - 1$	common output length	Part I’s $N = 2r - 1$ or $2m - 1$, also an output length, of other constructions
$C_{T,g}, \mathcal{W}_{T,g}$	weak compositions of T into g parts; the binary words with T letters a and $g - 1$ letters b	—
$\omega(\mathbf{c})$ (renamed from $w(\mathbf{c})$)	the binary word with gap vector $\mathbf{c}$, (11)	a word w ; w still names an arbitrary output word
$\mathcal{E}, \mathcal{E}_{\text{safe}}$ (renamed from E, E_{safe})	a set of target gap vectors; the safe vectors (23)	Part I’s E , the bad adjacent pairs of (9), and its nonterminal E_0
η_m (renamed from E_m)	$\max(0, m^2 - 6m + 3)$, (25)	—
$\Lambda = 2m + 1$ (renamed from D)	spacing of the centres, (27)	the dynamic-programming table $D(i, j, s)$ of Section 9 and the spectrum $\mathcal{D}(A, B)$
$\Delta(\mathbf{x}, \mathbf{y})$	half the ℓ^1 distance of equal-sum vectors, (19)	—
$\rho_j(\mathbf{v})$ (renamed from R_j)	the escape sums of (20)	the relation R of Lemma 8.1
$\mathbf{b}_m, b_{m,j}$	the base vector (25) and its coordinates	the letter b
Box_i (renamed from Q_i)	the target vectors below $\mathbf{v}_i$	$Q = \Sigma $ and the formulas $Q_{w,k}$ of Part I
κ_m (renamed from K_m)	$ \text{Box}_i $, (30)	the uniform degree bound K of Section 8
$h_{m,g,r}, H_m(z)$	singleton degree counts and their polynomial	—
Φ_n (renamed from J_n)	the Rodrigues polynomial (41)	Part I’s position sets $J_{m,t}$ of (7)
$\text{Deg}_{A,B}(z)$ (renamed from $F_{A,B}$)	the degree-enumerating polynomial (34)	Part I’s set F of forbidden triples, (3)
μ (renamed from M)	$\binom{N}{m-1} - 1$ in (44)	Part I’s M , the length-based maximum degree of its search encoding
p (renamed from Q)	a test polynomial in Lemma 18.1	$Q = \Sigma $
m^2	the minimum output length in the full-coverage class of Theorem 12.3	not a lower bound for every word of degree m : that bound is $2m - 1$

the inserted language $\{a^m\}$, and pays in length and in target words: for $q = 1$ and $m \geq 6$ its outputs have length exactly m^2 , and Theorem 12.3 shows that no single-unary-source pair whose shuffle is a whole binary fixed-letter-count class does better. The two statements do not conflict, because Part I's full coverage is of Σ^{2m-1} over a large alphabet with a many-word inserted language. For the spectrum $\{1, 2, 4\}$, Part I's binary example (10) uses six inserted and three target words of lengths four and three; Part II's construction for the list $(2, 4)$ uses one inserted word and 2570 target words, with outputs of length 30 (the row $m = 4$, profile 2; 4 of data/02-binary-profiles-profile_checks.csv). Neither source claims minimal language sizes.

12 Definitions and main statements

The oriented insertion degree is that of Section 2.

12.1 Fixed-letter-count classes

Write $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and, for $g \geq 1$, set

$$C_{T,g} = \{\mathbf{c} = (c_1, \dots, c_g) \in \mathbb{N}_0^g : c_1 + \dots + c_g = T\}.$$

Encode a gap vector by

$$\omega(\mathbf{c}) = a^{c_1} b a^{c_2} b \dots b a^{c_g}, \quad \mathcal{W}_{T,g} = \{\omega(\mathbf{c}) : \mathbf{c} \in C_{T,g}\}. \quad (11)$$

There are exactly $g - 1$ separators, even when some gap lengths are zero. Consequently the encoding is one-to-one, and $\mathcal{W}_{T,g}$ is the entire set of binary words with T copies of a and $g - 1$ copies of b . In particular,

$$|\mathcal{W}_{T,g}| = \binom{T + g - 1}{g - 1}. \quad (12)$$

This is often called a Parikh class; only the explicit description above is needed in the proofs.

Theorem 12.1 (Arbitrary binary profiles with one unary inserted word). *Let $m \geq 2$, and let $(t_0, \dots, t_{q-1})$ be a nonempty finite list with $2 \leq t_i \leq m$. There are an integer T , a finite language $B \subseteq \mathcal{W}_{T-m,m}$, and distinct words $W_0, \dots, W_{q-1}$ in $\mathcal{W}_{T,m}$ such that, for $A = \{a^m\}$,*

$$\mathcal{I}_k(A, B) = \mathcal{W}_{T,m} \setminus \{W_i : t_i > k\} \quad (k \geq 1). \quad (13)$$

In particular $A \sqcup B = \mathcal{W}_{T,m}$, $d_{A,B}(W_i) = t_i$, and every other output has degree 1. Each W_i is produced by exactly one target word of B , although it can have more than one position mask for that target.

An explicit formula for T and every word of B is given in Section 15. Repeated entries in the list are allowed and control multiplicities, not merely the support of the spectrum.

Corollary 12.2 (Fixed-binary classification). *The spectra of pairs of nonempty finite binary languages are exactly the finite subsets of $\mathbb{N}$ containing 1. Every such set is attained with the inserted language a singleton $\{a^m\}$. If the spectrum contains a degree greater than 1, two is its minimum possible alphabet size, and one is the minimum possible cardinality of the nonempty inserted language.*

Proof. A shuffle of finite languages is finite. Concatenating any chosen source pair shows that degree 1 occurs. Conversely, list the elements of $S \setminus \{1\}$ and apply Theorem 12.1 with $m = \max S$. For $S = \{1\}$ use $A = \{a\}$, $B = \{b\}$. Over a one-letter alphabet every shuffle output is the concatenation of its sources, so every degree is 1 (Proposition 2.3(iv)). Finally, a nonempty language cannot have fewer than one word. $\square$

Allowing empty input languages adds precisely the empty spectrum, as in Part I. In particular, for every $r \geq 2$ the least alphabet realizing $\{1, r\}$ has two letters, which answers the least-alphabet question of Part I's Section 10; the smallest-source question there is answered for the inserted language only (one word), not for the target language (Question 21.2).

Theorem 12.3 (Sharp isolated-witness length in the full-coverage class). *Let $m \geq 3$. Suppose $A = \{a^m\}$, $B \subseteq \mathcal{W}_{T-m,g}$ is nonempty, and*

$$A \sqcup B = \mathcal{W}_{T,g}.$$

If exactly one output has degree greater than 1, and its degree is m , then every output has length

$$N = T + g - 1 \geq m^2. \quad (14)$$

For every $m \geq 6$, equality is attained with $g = m$ and $T = m^2 - m + 1$. Thus the minimum length in this class is exactly m^2 for all $m \geq 6$.

The full-coverage and singleton-unary-source hypotheses are part of the extremal theorem. The assertion is *not* a universal m^2 lower bound for a word of insertion degree m : without these hypotheses the elementary length bound is only $2m - 1$.

13 Gap vectors and the minimum over representations

For vectors, inequalities are coordinatewise, and $\text{supp}(\mathbf{z}) = \{j : z_j \neq 0\}$.

Lemma 13.1 (Gap-support formula). *Fix $m \geq 1$, $g \geq 1$, and $\beta \in \mathbb{N}_0^g$. A word $\omega(\mathbf{c})$ is a shuffle of a^m and $\omega(\beta)$ if and only if*

$$\beta \leq \mathbf{c}, \quad \sum_j (c_j - \beta_j) = m.$$

For every such output,

$$d_{\{a^m\}, \{\omega(\beta)\}}(\omega(\mathbf{c})) = |\text{supp}(\mathbf{c} - \beta)|. \quad (15)$$

Proof. The inserted source supplies no b . Hence all b 's come from the target, in their original order. The j th output gap retains the target's β_j copies of a and receives exactly $c_j - \beta_j$ new copies. This proves the necessary coordinate and sum conditions; they are sufficient by placing the new copies in those gaps.

Each gap receiving new copies contains at least one inserted-source run. Distinct gaps are separated by a target b , so those runs cannot merge. Conversely, within each receiving gap put all its new copies consecutively. This gives exactly one run per receiving gap and proves the minimum formula. The argument minimizes over all masks for the fixed target; it does not assume that a mask is unique. $\square$

Corollary 13.2 (Language-level gap optimization). *Let $\mathcal{E} \subseteq C_{T-m,g}$ and $B = \{\omega(\beta) : \beta \in \mathcal{E}\}$. For $\mathbf{c} \in C_{T,g}$,*

$$d_{\{a^m\}, B}(\omega(\mathbf{c})) = \min_{\substack{\beta \in \mathcal{E} \\ \beta \leq \mathbf{c}}} |\text{supp}(\mathbf{c} - \beta)|, \quad (16)$$

where an empty minimum means the word is not a shuffle output. In particular, it has degree 1 precisely when

$$\mathbf{c} - m\mathbf{e}_j \in \mathcal{E} \quad \text{for some } j \text{ with } c_j \geq m. \quad (17)$$

Proof. Take the minimum of (15) over the target words. A nonnegative vector of total m has support one exactly when it is $m\mathbf{e}_j$. $\square$

The distinction between (15) and (16) is the entire source of the examples below. A union of target languages takes a pointwise minimum of costs, not a union of the sets of their minimum costs.

13.1 The exact singleton histogram

Proposition 13.3 (Singleton degree counts). *For every fixed target gap vector $\beta \in \mathbb{N}_0^g$, the number of distinct outputs of the pair $(a^m, \omega(\beta))$ whose minimum degree is r is*

$$h_{m,g,r} = \binom{g}{r} \binom{m-1}{r-1} \quad (1 \leq r \leq \min(m, g)). \quad (18)$$

These counts are independent of β . The total number of outputs is $\binom{m+g-1}{g-1}$, and the spectrum is $\{1, \dots, \min(m, g)\}$.

Proof. Every output corresponds to a unique weak composition $\delta = \mathbf{c} - \beta$ of m into g parts. To have support r , choose its r positive coordinates in $\binom{g}{r}$ ways and distribute m positively over them in $\binom{m-1}{r-1}$ ways. The encoding by gap vectors is injective, so these count distinct words rather than masks. Summing counts all weak compositions of m . $\square$

The spectrum conclusion is Hughes's unary-source singleton theorem [1, Theorem 9]; the displayed counts are its elementary counting refinement. The same proof works with an arbitrary fixed sequence of non- a separator letters, not only repeated b 's.

14 A geometric realization principle

We first isolate the geometry needed to protect exceptional outputs. This also makes the construction adaptable to other parameter choices. For equal-sum vectors, define

$$\Delta(\mathbf{x}, \mathbf{y}) = \sum_j (x_j - y_j)_+ = \frac{1}{2} \sum_j |x_j - y_j|. \quad (19)$$

The second equality follows by equating total positive and negative parts. Consequently Δ is a metric on each fixed-sum class.

Lemma 14.1 (A target below an exceptional vector is nearby). *Let $\mathbf{c}, \mathbf{v} \in C_{T,g}$ and $\gamma \in \mathbb{N}_0^g$ have total m . If $\mathbf{c} - \gamma \geq 0$ and $\mathbf{c} - \gamma \leq \mathbf{v}$, then*

$$\Delta(\mathbf{c}, \mathbf{v}) \leq m.$$

Proof. For each coordinate, $(c_j - v_j)_+ \leq \gamma_j$. Sum. $\square$

Lemma 14.2 (One-piece escape). *Let $\mathbf{v} \in C_{T,g}$ and assume*

$$T > g(m-1), \quad \rho_j(\mathbf{v}) := \sum_{k \neq j} (v_k - m + 1)_+ > m \quad (1 \leq j \leq g). \quad (20)$$

For every $\mathbf{c} \in C_{T,g}$ with $\mathbf{c} \neq \mathbf{v}$, there exists j such that $c_j \geq m$ and

$$\mathbf{c} - m\mathbf{e}_j \not\leq \mathbf{v}. \quad (21)$$

Proof. Call j eligible if $c_j \geq m$. At least one coordinate is eligible, otherwise $T \leq g(m-1)$. Also some coordinate i satisfies $c_i > v_i$, because the two vectors have the same total and are distinct.

If at least two coordinates are eligible, choose an eligible $j \neq i$. Subtracting m there leaves the excess at i , proving (21). Suppose instead that j is the unique eligible coordinate. If an excess $c_i > v_i$ occurs at an $i \neq j$, subtraction at j again preserves it. The remaining case has

$$c_k \leq \min(v_k, m - 1) \quad (k \neq j).$$

Equality of totals then gives

$$c_j - v_j = \sum_{k \neq j} (v_k - c_k) \geq \sum_{k \neq j} (v_k - m + 1)_+ = \rho_j(\mathbf{v}) > m.$$

Thus $c_j - m > v_j$, and subtracting at j still leaves an excess. These cases exhaust all possibilities. $\square$

Remark 14.3 (Why both hypotheses matter). Large values of ρ_j do not by themselves imply that an eligible coordinate exists. For $m = g = 6$, the vector $(12, 12, 1, 1, 1, 1)$ satisfies $\rho_j > 6$ for every j , but has total 28. The other vector $(5, 5, 5, 5, 4, 4)$ has the same total and no coordinate at least 6. It admits no one-piece insertion of a^6 at all. The strict total-size condition in (20) cannot be dropped. This obstruction is the reason the sharp construction becomes quadratic rather than remaining linear in m .

Theorem 14.4 (Separated exceptional vectors). *Let $m \geq 1$, $g \geq 2$, $q \geq 1$, and let distinct vectors $\mathbf{v}_0, \dots, \mathbf{v}_{q-1} \in C_{T,g}$ satisfy (20) and*

$$\Delta(\mathbf{v}_i, \mathbf{v}_h) > 2m \quad (i \neq h). \quad (22)$$

Choose $\delta_i \leq \mathbf{v}_i$ of total m , with $t_i = |\text{supp}(\delta_i)| \geq 2$, and put $\beta_i = \mathbf{v}_i - \delta_i$. Define

$$\mathcal{E}_{\text{safe}} = \{\beta \in C_{T-m,g} : \beta \not\leq \mathbf{v}_i \text{ for every } i\}, \quad (23)$$

$$\mathcal{E} = \mathcal{E}_{\text{safe}} \cup \{\beta_0, \dots, \beta_{q-1}\}, \quad B = \{\omega(\beta) : \beta \in \mathcal{E}\}. \quad (24)$$

Then $A = \{a^m\}$ and B satisfy

$$\mathcal{I}_k(A, B) = \mathcal{W}_{T,g} \setminus \{\omega(\mathbf{v}_i) : t_i > k\} \quad (k \geq 1).$$

Each exceptional output $\omega(\mathbf{v}_i)$ has the unique target $\omega(\beta_i)$ in B .

Proof. All possible shuffles have the asserted letter counts. We determine the minimum degree of every vector in $C_{T,g}$.

Exceptional vectors. No element of $\mathcal{E}_{\text{safe}}$ lies below $\mathbf{v}_i$. For $h \neq i$, the inequality $\beta_h \leq \mathbf{v}_i$, together with $\mathbf{v}_h - \beta_h = \delta_h$ of total m , would imply $\Delta(\mathbf{v}_h, \mathbf{v}_i) \leq m$ by Lemma 14.1. This contradicts separation. Thus the only accepted target below $\mathbf{v}_i$ is β_i , and Lemma 13.1 gives minimum degree t_i . The same argument shows that the special targets are distinct and proves uniqueness of the target word.

Every other vector. Fix $\mathbf{c}$ distinct from all exceptional vectors. A center $\mathbf{v}_i$ will be called an obstruction for $\mathbf{c}$ if some eligible coordinate j has $\mathbf{c} - m\mathbf{e}_j \leq \mathbf{v}_i$. By Lemma 14.1, each such center lies at distance at most m from $\mathbf{c}$. There is at most one obstructing center: two would violate (22) by the triangle inequality.

There is at least one eligible coordinate because $T > g(m - 1)$. If no center is an obstruction, any eligible subtraction lies in $\mathcal{E}_{\text{safe}}$. If $\mathbf{v}_i$ is the sole obstruction, Lemma 14.2 gives an eligible subtraction not below that center. It cannot be below any other center, because that would make the other center an obstruction as well. Again it lies in $\mathcal{E}_{\text{safe}}$. In both cases Corollary 13.2 gives degree 1.

Every vector has now been assigned an accepted representation, and its minimum degree has been proved. This establishes the entire hierarchy. $\square$

15 Explicit binary profiles and exact sizes

The next parameter choices turn Theorem 14.4 into a completely explicit recipe. They keep the alphabet fixed and give the sharp isolated length for all sufficiently large m .

15.1 Base vectors and separated translates

For $m \geq 2$, put $\Lambda = 2m + 1$ and define a vector $\mathbf{b}_m$ with m coordinates by

$$\mathbf{b}_m = \begin{cases} (4, 4), & m = 2, \\ (4, 4, 4), & m = 3, \\ (2m + \eta_m, 2m, 1, \dots, 1), & m \geq 4, \end{cases} \quad \eta_m = \max(0, m^2 - 6m + 3). \quad (25)$$

Let $T_0 = \sum_j b_{m,j}$. Thus

$$T_0 = \begin{cases} 8, & m = 2, \\ 12, & m = 3, \\ \max(5m - 2, m^2 - m + 1), & m \geq 4. \end{cases} \quad (26)$$

Given a profile list $(t_0, \dots, t_{q-1})$, set

$$\mathbf{v}_i = \mathbf{b}_m + \Lambda i \mathbf{e}_1 + \Lambda(q - 1 - i) \mathbf{e}_2, \quad 0 \leq i < q, \quad (27)$$

$$T = T_0 + \Lambda(q - 1), \quad N = T + m - 1, \quad (28)$$

$$\delta_i = (m - t_i + 1, \underbrace{1, \dots, 1}_{t_i - 1}, \underbrace{0, \dots, 0}_{m - t_i}), \quad \beta_i = \mathbf{v}_i - \delta_i. \quad (29)$$

Use (23)–(24) with these vectors. Equations (25)–(29) and the membership condition are an explicit specification of every target word; no existential choices remain.

Lemma 15.1 (The parameters meet every hypothesis). *The vectors in (27) and (29) satisfy all the hypotheses of Theorem 14.4, with $g = m$.*

Proof. Every center has total T and positive coordinates. For $m = 2$, $T_0 = 8 > 2$ and each ρ_j of the base equals $3 > 2$. For $m = 3$, $T_0 = 12 > 6$ and each ρ_j equals $2(4 - 3 + 1) = 4 > 3$.

For $m \geq 4$, $T_0 \geq m(m - 1) + 1$. The first two base coordinates are at least $2m$. After any one coordinate is excluded from the sum defining ρ_j , at least one of these two remains. It contributes at least $2m - m + 1 = m + 1 > m$. Increasing coordinates preserves all these inequalities, so every translated center satisfies the escape hypotheses.

For distinct indices,

$$\Delta(\mathbf{v}_i, \mathbf{v}_h) = \Lambda|i - h| > 2m.$$

Each deficit has total m and exactly t_i positive coordinates. Its first coordinate is at most $m - 1$, while the first coordinate of the center is at least m . Every other positive deficit coordinate is one and every center coordinate is positive. Thus $\delta_i \leq \mathbf{v}_i$. $\square$

Proof of Theorem 12.1. Apply Theorem 14.4 to Lemma 15.1, taking $W_i = \omega(\mathbf{v}_i)$. $\square$

15.2 Exact language cardinalities

Proposition 15.2 (Input and output sizes). *For the explicit construction, define*

$$\kappa_m = \begin{cases} 3, & m = 2, \\ 10, & m = 3, \\ (m + 4)2^{m-3}, & m \geq 4. \end{cases} \quad (30)$$

Then

$$|A| = 1, \quad |B| = \binom{T-1}{m-1} - q(\kappa_m - 1), \quad (31)$$

$$|A \sqcup B| = \binom{N}{m-1}, \quad |\mathcal{I}_k(A, B)| = \binom{N}{m-1} - |\{i : t_i > k\}|. \quad (32)$$

Every target word has length $N - m$, and every output has length N .

Proof. At total $T - m$, let $\text{Box}_i = \{\boldsymbol{\beta} \in C_{T-m,m} : \boldsymbol{\beta} \leq \mathbf{v}_i\}$. These sets are pairwise disjoint. Indeed, if $\boldsymbol{\beta} \leq \mathbf{v}_i, \mathbf{v}_h$, then on the coordinates where $v_{i,j} > v_{h,j}$ one has $v_{i,j} - v_{h,j} \leq v_{i,j} - \beta_j$. Summing bounds $\Delta(\mathbf{v}_i, \mathbf{v}_h)$ by m , contradicting separation.

Count Box_i through deficits $\boldsymbol{\delta} = \mathbf{v}_i - \boldsymbol{\beta}$ of total m . For $m = 2$ and $m = 3$ none of their coordinates is capped below m , so the counts are $\binom{3}{1} = 3$ and $\binom{5}{2} = 10$. For $m \geq 4$, the first two deficit coordinates are unrestricted, while the remaining $m - 2$ lie in $\{0, 1\}$. Hence

$$|\text{Box}_i| = \sum_{s=0}^{m-2} \binom{m-2}{s} (m-s+1) = (m+1)2^{m-2} - (m-2)2^{m-3} = \kappa_m.$$

The safe set removes all $q\kappa_m$ vectors and the construction restores one in each box, proving (31). The exceptional targets are distinct and were all removed before restoration. The output formulas follow from Theorem 12.1 and (12). Letter counts give the length statements. $\square$

For one prescribed degree m , representative sizes are as follows. No minimality is asserted in the first four rows.

m	T_0	output length N	$ B $	$ A \sqcup B $
2	8	9	5	9
3	12	14	46	91
4	18	21	665	1,330
5	23	27	7,280	17,550
6	31	36	142,427	376,992

For $m \geq 6$, the length for an arbitrary q -entry profile is

$$N = m^2 + (2m + 1)(q - 1). \quad (33)$$

The exact $m = 5$ count in the table is $\binom{22}{4} - (36 - 1) = 7315 - 35 = 7280$.

15.3 All possible finite higher-degree count sequences

Corollary 15.3 (Arbitrary higher-degree multiplicities). *Fix $m \geq 2$ and nonnegative integers $q_2, \dots, q_m$, not all zero. There are finite binary languages as above for which*

$$|\{w : d_{A,B}(w) = r\}| = q_r \quad (2 \leq r \leq m),$$

and every other output has degree 1. Consequently the degree-enumerating polynomial is

$$\text{Deg}_{A,B}(z) = \left(\binom{N}{m-1} - q \right) z + \sum_{r=2}^m q_r z^r, \quad q = \sum_{r=2}^m q_r. \quad (34)$$

Proof. Repeat the entry r exactly q_r times in the profile list and apply Theorem 12.1 and Proposition 15.2. $\square$

The degree-one coefficient is determined by the construction. The theorem does not prescribe that coefficient independently. The entirely zero higher-degree profile is already represented by any nonempty pair with spectrum $\{1\}$.

16 Sharp quadratic size for an isolated high degree

The optimized parameter choice is forced by a combinatorial obstruction, not merely suggested by finite experiments.

16.1 Outputs with no long enough gap

Lemma 16.1 (A capped-composition obstruction). *Assume $A = \{a^m\}$, $B \subseteq \mathcal{W}_{T-m,g}$, and $A \sqcup B = \mathcal{W}_{T,g}$. The number of outputs with degree greater than 1 is at least*

$$[x^T](1 + x + \dots + x^{m-1})^g. \quad (35)$$

If $m \geq 2$ and $0 < T < g(m-1)$, this coefficient is at least g .

Proof. Each vector with every coordinate at most $m-1$ lacks an eligible one-run deletion in (17). Full coverage guarantees it is nevertheless an output, so its degree exceeds one. The coefficient counts exactly those vectors.

For the second assertion, the case $g = 1$ has coefficient one. For $g \geq 2$, form a vector by filling coordinates greedily to $m-1$, then using a possible remainder, then zeros. At an interior total it is not constant. Permuting its coordinates yields at least g distinct vectors: if a value has multiplicity s with $1 \leq s < g$, the number of permutations is at least $\binom{g}{s} \geq g$. Each permutation still has the required total and coordinate cap. $\square$

16.2 The boundary total cannot attain high degree

Proposition 16.2 (Universal lower bound in the specified class). *Under the hypotheses of Theorem 12.3, one has*

$$g \geq m, \quad T \geq g(m-1) + 1, \quad N \geq gm \geq m^2.$$

Proof. An inserted word of length m can occupy at most g gaps, so an output of degree m forces $g \geq m$. Also $T \geq m$ because the nonempty target language contains $T - m$ copies of a .

If $T < g(m - 1)$, Lemma 16.1 supplies at least $g \geq m \geq 3$ outputs of degree greater than 1, contrary to uniqueness. Thus $T \geq g(m - 1)$. Suppose equality holds. The vector

$$\mathbf{v}_* = (m - 1, m - 1, \dots, m - 1)$$

has no eligible one-run deletion, so it must be the unique exceptional vector. Consider its distinct neighbor

$$\mathbf{c} = (m, m - 2, m - 1, \dots, m - 1).$$

It has the same total, hence is an output, and must have degree 1. Only its first gap has length at least m . Its only possible one-run target is therefore

$$\boldsymbol{\beta} = (0, m - 2, m - 1, \dots, m - 1),$$

which must be in the target set. But $\boldsymbol{\beta} \leq \mathbf{v}_*$ and

$$\mathbf{v}_* - \boldsymbol{\beta} = (m - 1, 1, 0, \dots, 0).$$

By Lemma 13.1, this target gives the exceptional output a representation of degree 2, contradicting its degree $m \geq 3$. Equality in the total is impossible. Finally $N = T + g - 1 \geq gm \geq m^2$. $\square$

16.3 An explicit optimal family

For $m \geq 6$, take $q = 1$ and $t_0 = m$. The center is

$$\mathbf{v} = (m^2 - 4m + 3, \underbrace{2m, 1, \dots, 1}_{m-2}), \quad T = m^2 - m + 1. \quad (36)$$

The first coordinate is at least $2m$ because $m^2 - 6m + 3 \geq 0$ for all integers $m \geq 6$. The special target gap vector is $\boldsymbol{\beta}_* = \mathbf{v} - (1, \dots, 1)$. In ordinary word notation,

$$W_* = a^{m^2-4m+3} b a^{2m} (ba)^{m-2}, \quad (37)$$

$$V_* = a^{m^2-4m+2} b a^{2m-1} b^{m-2}. \quad (38)$$

Let B contain V_* and every target word whose gap vector has total $(m - 1)^2$ and is not coordinatewise below $\mathbf{v}$. Then W_* is the unique degree- m output, all other words with T copies of a and $m - 1$ copies of b have degree 1, and the common output length is m^2 .

Proof of Theorem 12.3. The lower bound is Proposition 16.2. Equations (36)–(38) are the $q = 1$ instance of Theorem 12.1, whose hypotheses were proved in Lemma 15.1. Its length is $T + m - 1 = m^2$, giving equality for every $m \geq 6$. $\square$

This theorem minimizes over *all* finite target languages in the stated full-coverage class, not only over targets arising from the separated-center construction. It also permits arbitrary numbers of separator letters in the lower bound. For $m \geq 6$, any equality case necessarily has $g = m$ and $T = m^2 - m + 1$, since both inequalities $N \geq gm \geq m^2$ must be equalities. No classification of the target languages at equality is claimed. [Added 30 September 2026, batch 66: Part III classifies the equality cases, for every $m \geq 3$ at the least length: the centers (Theorems 29.1 and 31.1), the mandatory targets (Theorem 33.2) and, by a finite covering criterion, all realizing target languages (Corollary 27.3). The covering problem is solved completely only for $m = 3$ (Theorem 34.1).]

17 A fully explicit fourteen-symbol gap

The balanced $m = 3$ instance is particularly easy to inspect. Set

$$A = \{aaa\}, \quad B = \{a^i ba^j ba^k : i + j + k = 9, \max(i, j, k) \geq 5\} \cup \{a^3 ba^3 ba^3\}. \quad (39)$$

There are $\binom{11}{2} = 55$ nonnegative triples of total 9. Exactly ten are coordinatewise at most $(4, 4, 4)$, since subtracting them from $(4, 4, 4)$ gives the ten weak compositions of 3 into three parts. Removing those ten and restoring $(3, 3, 3)$ gives $|B| = 46$.

Theorem 17.1 (The worked example). *For (39), with $W = a^4 ba^4 ba^4$,*

$$\mathcal{I}_1(A, B) = \mathcal{I}_2(A, B) = \mathcal{W}_{12,3} \setminus \{W\}, \quad \mathcal{I}_k(A, B) = \mathcal{W}_{12,3} \quad (k \geq 3).$$

There are 90 degree-one outputs and one degree-three output. Every one of the 46 singleton source pairs has exactly three degree-one, six degree-two, and one degree-three outputs.

Proof. For W , any valid target is coordinatewise at most $(4, 4, 4)$. Among the accepted targets, only $(3, 3, 3)$ has that property. Its deficit is $(1, 1, 1)$, which forces degree three.

For any other vector $\mathbf{c}$ of total 12, the escape lemma applies to $\mathbf{v} = (4, 4, 4)$ with $m = g = 3$: the total exceeds $3(3 - 1)$, and each ρ_j is $4 > 3$. It gives a coordinate with $c_j \geq 3$ for which $\mathbf{c} - 3\mathbf{e}_j \not\leq (4, 4, 4)$. That target has a coordinate at least five and belongs to B , giving degree one. Thus all $\binom{14}{2} = 91$ output words occur with the stated degrees. The singleton counts are (18) at $m = g = 3$. $\square$

For example, the output with vector $(5, 4, 3)$ is easy: remove all three inserted a 's from its second gap to obtain the target $(5, 1, 3)$. On the other hand, the special target $(3, 3, 3)$ has the ten outputs below. Only one retains its higher degree when the remaining 45 target words are admitted.

Deficit type	Output gap vectors	Singleton degree
$(3, 0, 0)$ and permutations	$(6, 3, 3)$ and permutations	1
$(2, 1, 0)$ and permutations	$(5, 4, 3)$ and permutations	2
$(1, 1, 1)$	$(4, 4, 4)$	3

Across all 46 pairs there are $46 \cdot 6 = 276$ pair/output incidences of singleton degree two. Every one is assigned degree one at the language level. This number counts incidences, not distinct words. The shipped files `data/02-binary-profiles-worked_targets.csv` and `data/02-binary-profiles-worked_outputs.csv` record all 46 target words and all 91 language-level output degrees.

18 Identical real-rooted singleton histograms, arbitrary aggregate gaps

There is a stronger algebraic contrast than the absence of a singleton gap. Every target in the $g = m$ construction has the same polynomial

$$H_m(z) = \sum_{r=1}^m \binom{m}{r} \binom{m-1}{r-1} z^r. \quad (40)$$

We give a short self-contained proof that all its zeros are real. This is a classical orthogonal-polynomial type argument; no priority is asserted for the real-rootedness fact itself. Its role here is to delimit what can survive minimization over source representations.

18.1 A Rodrigues polynomial

For $n \geq 0$, define

$$\Phi_n(x) = \frac{1}{n!x} \frac{d^n}{dx^n} [x^{n+1}(1-x)^n]. \quad (41)$$

The apparent singularity at zero is removable: every differentiated monomial still contains a factor x . Expansion gives

$$\Phi_n(x) = \sum_{j=0}^n (-1)^j \binom{n}{j} \binom{n+j+1}{n} x^j, \quad (42)$$

so it has degree exactly n .

Lemma 18.1. *For $n \geq 1$, Φ_n has n distinct zeros, all in $(0, 1)$.*

Proof. For every polynomial p of degree less than n , integration by parts n times gives

$$\int_0^1 x \Phi_n(x) p(x) dx = \frac{1}{n!} \int_0^1 \frac{d^n}{dx^n} [x^{n+1}(1-x)^n] p(x) dx = 0.$$

Every boundary term vanishes, because derivatives of order at most $n-1$ of $x^{n+1}(1-x)^n$ vanish at both endpoints. The remaining integral contains $p^{(n)} = 0$.

Suppose Φ_n had fewer than n sign-changing zeros in $(0, 1)$. Let p be the product of $x - r$ over those zeros, once each. Its degree is less than n , while $\Phi_n p$ has one constant sign on $(0, 1)$ apart from its finitely many zeros and is not identically zero. Multiplying by $x > 0$ and integrating cannot give zero, a contradiction. Thus there are at least n sign-changing zeros there. Degree n forces exactly n , all simple. $\square$

Lemma 18.2 (Finite polynomial identity). *For $n = m - 1$,*

$$H_m(z) = z(1-z)^n \Phi_n\left(\frac{z}{z-1}\right). \quad (43)$$

The right side is interpreted after polynomial cancellation at $z = 1$.

Proof. Using (42), the coefficient of z^k in $(1-z)^n \Phi_n(z/(z-1))$ is

$$\sum_{j=0}^k (-1)^{k-j} \binom{n}{j} \binom{n+j+1}{n} \binom{n-j}{k-j}.$$

Since $\binom{n}{j} \binom{n-j}{k-j} = \binom{n}{k} \binom{k}{j}$, this becomes

$$\binom{n}{k} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} \binom{n+j+1}{n} = \binom{n}{k} \binom{n+1}{k+1}.$$

For the last equality, apply the forward-difference identity $\binom{y+1}{n} - \binom{y}{n} = \binom{y}{n-1}$ successively k times, starting at $y = n+1$. These are exactly the coefficients of $H_m(z)/z$ in (40), proving the identity. $\square$

Theorem 18.3 (Real-rooted singleton histograms). *For $m \geq 2$, H_m has a simple zero at 0 and $m-1$ distinct negative real zeros.*

Proof. Each zero $x \in (0, 1)$ of Φ_{m-1} gives the distinct negative zero $z = x/(x-1)$ in (43). The additional zero at 0 is simple because $[z]H_m = m$. There are m zeros in total, so these exhaust its degree. $\square$

18.2 The aggregate can have almost no real zeros

For an isolated degree m , (34) reduces to

$$\text{Deg}_{A,B}(z) = \mu z + z^m, \quad \mu = \binom{N}{m-1} - 1 > 0. \quad (44)$$

Apart from zero, its roots solve $z^{m-1} = -\mu$. If m is odd, all $m-1$ nonzero roots are nonreal. If m is even, exactly one is negative real and $m-2$ are nonreal. Thus the proportion of nonreal roots tends to one as $m \rightarrow \infty$, even though *each* singleton source-pair polynomial is exactly the same real-rooted polynomial H_m .

Already $m = 3$ gives the coefficient sequence $(\mu, 0, 1)$, which fails log-concavity. Thus neither a general log-concavity nor a no-internal-zero conclusion survives this minimization. This is not a counterexample to a theorem about sums of real-rooted polynomials: the language-level polynomial is not their sum. It counts each output once at its cheapest degree, an operation that retains neither the singleton coefficient sequence nor its real zeros.

19 Algorithms, succinct descriptions, and certificates

19.1 Membership without enumerating the target language

Although $|B|$ can be large, the vector description is short. A target word is parsed into its gap vector β , checked to have $m-1$ separators and total $T-m$, and then accepted exactly if

$$\beta \in \{\beta_i : 0 \leq i < q\} \quad \text{or} \quad \forall i \quad \beta \not\leq \mathbf{v}_i.$$

A direct implementation uses $O(N + qm)$ elementary operations on the explicit word and the stored vectors. This is a membership-algorithm bound, not a claim that a minimal deterministic finite automaton has that size.

The safe-set test has an even simpler form for the translates (27). If some $j \geq 3$ has $\beta_j > b_{m,j}$, the vector is safe. Otherwise, domination by some center is equivalent to the existence of an integer i in the interval

$$\max \left(0, \left\lceil \frac{\beta_1 - b_{m,1}}{\Lambda} \right\rceil \right) \leq i \leq \min \left(q-1, q-1 - \left\lceil \frac{\beta_2 - b_{m,2}}{\Lambda} \right\rceil \right). \quad (45)$$

The interval is empty precisely when the vector is safe. This follows by solving the first two coordinate inequalities for i ; the remaining coordinate inequalities have already been checked. For $m = 2$ there are no remaining coordinates. Ceiling division must retain its mathematical meaning for negative numerators.

All target languages are finite and hence regular: a trie of their words is one explicit finite automaton. The concise membership test should not be confused with a minimal-state construction, and no unproved polynomial DFA state bound is used.

19.2 Computing a minimum degree

For an arbitrary finite gap-vector target set, (16) is an exact algorithm. An alternative enumerates weak compositions δ of m and tests whether $\delta \leq \mathbf{c}$ and $\mathbf{c} - \delta \in \mathcal{E}$. Ordering the compositions by support size permits stopping at the first accepted one. For $g = m$, at most $\binom{2m-1}{m-1}$ candidates are tested.

For the constructed languages there is a faster certified route. Compare the output vector with the q centers. Equality with a center gives its declared degree, with the uniqueness proof certifying minimality. For any

other output, try the at most m vectors $\mathbf{c} - m\mathbf{e}_j$ with $c_j \geq m$. The escape argument guarantees that one is safe. Direct comparisons give an $O(qm^2)$ arithmetic-operation bound, in addition to reading the word. The successful target and chosen gap are an explicit degree-one certificate.

19.3 A literal-word dynamic program

For independent verification, it is useful to avoid the gap lemma entirely. Fix literal words u, v, w with $|w| = |u| + |v|$. A state (i, j, ℓ) records how many letters of u and v have been consumed and whether the preceding output letter was assigned to u ($\ell = 1$) or not ($\ell = 0$). Initialize $(0, 0, 0)$ at cost zero. At output position $i + j$, a transition taking the next matching letter of u adds $1 - \ell$ to the cost and sets $\ell = 1$; a transition taking the next matching letter of v adds zero and sets $\ell = 0$. Keep the smallest cost at every state.

Induction on the consumed output prefix shows that the state cost is exactly the smallest number of inserted-source runs among its compatible partial masks. Hence the minimum terminal cost is the desired degree for nonempty u . There are $O((|u| + 1)(|v| + 1))$ states and constant work per outgoing transition. This program can be compared directly with enumeration of the $\binom{|u|+|v|}{|u|}$ position masks, providing a check by a different algorithm. [Merge note: this is the dynamic program of Section 9, with $\ell = 1$ for Part I's state $s = A$ and $\ell = 0$ for $s = B$; Part I proves its exactness in the unnumbered proposition there.]

20 Executed validation and proof audit

The infinite families are proved in the preceding sections. Computations are additional exact checks, not inductive evidence substituted for proofs. The standard-library script, shipped as `code/02-binary-profiles-verify.py` (delivered as `code/verify.py`), was executed with Python 3.13.5; it uses explicit exceptions, so validation is not disabled by Python's optimization flag. Its executed receipt is `data/02-binary-profiles-verification.json`.

Executed check	Count
Complete constructed profiles	26
Output/profile comparisons, exhaustive in those profiles	493,704
Direct/interval target-membership comparisons	207,694
Literal position masks for the worked example	16,744
Independent fixed-pair/output dynamic-program comparisons	4,186
Weak-composition singleton histograms	56
Exact coefficients in the Rodrigues identity	819
Large-parameter hypothesis checks, up to $m = 80$	316
Seeded large-parameter output samples	12,640
Capped-composition coefficient levels	315
Boundary obstruction checks by literal-word dynamic programming	15

The exhaustive profile checks include every list over $\{2, 3\}$ of lengths one through three, several mixed degree-four and degree-five profiles, and all 376,992 outputs of the optimal degree-six example. For each output the script computes the minimum by cost-ordered exhaustive deficit enumeration against the *definition* of the target set, and compares that minimum with the theorem. Its early stop at cost one is exact because no smaller positive degree exists. The sizes of all target sets in these 26 cases are also counted independently by enumerating their gap vectors.

For the worked example, literal masks and the literal-word dynamic program agree for every source pair and every word of the correct letter counts, including the fixed-pair nonmembers. All 91 language-level output degrees and all 46 targets are recorded. The larger $m \leq 80$ output tests are explicitly *samples*, not exhaustive coverage. Their parameters and random seed (20260928) are stored in the JSON receipt. A regression guardrail rejects the insufficient-total example of Remark 14.3.

[Added in the merge: the suite was rerun on a copy on 29 September 2026 with Python 3.14.4, writing to a scratch directory. It passed; its three CSV files were byte-identical to the shipped ones, and its JSON receipt equal to the shipped one in every field except `python_version`.]

20.1 Specific failure modes excluded by the proofs

The proof does not rely on a uniquely chosen position mask; only the target word for an exceptional output is unique. It does not transfer an interval property through a union of targets; it calculates the resulting minimum. It does not assume that a fixed-sum output has an eligible one-run deletion: that is guaranteed by the separate strict total-size condition. It does not infer infinite validity from the enumerated cases. Finally, the optimal-length lower bound is proved for its stated full-coverage class and is not extended to arbitrary binary spectra.

The finite histogram identity is checked through degree 40, but its root location comes from the integration-by-parts proof, not from floating-point root plots. No external theorem-proving service, SMT solver, or network data is needed to run the verification suite. No Lean proof was produced for Part II.

21 Further research questions and topics

The following program separates the cited both-singleton question from new questions suggested by the present construction. Except where a source is explicitly named, these are proposed research problems rather than claims about an exhaustive literature consensus. Together with the questions of Part I's Section 10 that Part II leaves open, they are the report's current open list.

Question 21.1 (The unrestricted both-singleton conjecture). Does $\mathcal{D}(\{u\}, \{v\})$ always form an initial interval for arbitrary words u, v ? This is Hughes's Conjecture 2 [1, Section 11]. The present construction leaves it untouched: it keeps $u = a^m$ fixed but uses multiple targets. A useful next case is a two-run inserted word such as $a^p b^q$, where the single-skeleton gap model no longer applies unchanged.

Question 21.2 (How small can the target language be?). For a prescribed spectrum, or prescribed higher-degree multiplicities, what is the minimum $|B|$ when A is a singleton? The inserted-language size here is optimal, but optimality of the target counts in (31) is not established. In particular, how many targets are needed to erase all intermediate degrees below a surviving degree m ? A lower bound must account for overlaps between the sets of outputs a single target makes cheaper. [Added 30 September 2026, batch 66: Part III answers a restricted case. For one isolated degree m in the full-coverage class, at the shortest length for $m = 3$, the minimum is $|B| = 37$ (Theorem 34.1), and Corollary 35.2 gives a general counting lower bound. The question stays open otherwise; Question 38.1 continues it.]

Question 21.3 (Small isolated full-coverage witnesses). Determine the exact minimum length in Theorem 12.3 for $m = 3, 4, 5$. The proved lower bounds are 9, 16, 25, while this construction gives upper bounds 14, 21, 27. Varying the number of gaps, rather than only the center within $g = m$ gaps, may matter. No optimality of the small examples is asserted here. [Added 30 September 2026, batch 66: answered in Part III. The minima are 14, 21 and 27, the upper bounds above (Theorem 30.3). More gaps never help: the least length with g gaps, (71), increases strictly with g .]

Question 21.4 (Classification at the sharp bound). For $m \geq 6$, classify the pairs attaining $N = m^2$. The lower-bound proof forces $g = m$ and $T = m^2 - m + 1$, but does not force the center shape (36). Which vectors permit one isolated degree- m output? Which target words are mandatory in every realizing language? [Added 30 September 2026, batch 66: answered in Part III, for every $m \geq 3$, as far as the centers and the mandatory targets go. The centers at the least length are exactly the positive m -tuples of total T_m satisfying (74) (Theorem 31.1; for $m = 6$ there are 19 permutation orbits, one of them that of (36)), and the mandatory targets are classified by Theorem 33.2. The realizing languages are characterized by Corollary 27.3 but determined completely only for $m = 3$ (Theorem 34.1); for $m \geq 4$ the least language remains open (Question 38.1).]

Question 21.5 (Optimal packing for many exceptions). For $q > 1$, can the added length $(2m + 1)(q - 1)$ in (33) be reduced by placing centers in more than two varying coordinates? The relevant objects are fixed-sum integer vectors whose radius- m obstruction neighborhoods do not overlap. Separation greater than $2m$ is sufficient; determining when overlapping neighborhoods can be handled without introducing unintended degrees is a sharper problem.

Question 21.6 (Removing full coverage). Without requiring $A \sqcup B$ to fill an entire fixed-letter-count class, how short can a binary witness to spectrum $\{1, m\}$ be with $A = \{a^m\}$? The general lower bound is $2m - 1$, while the present family uses m^2 for $m \geq 6$. The quadratic lower bound proved here does not answer this question. A construction of length $O(m)$ would require avoiding the many short-gap words, rather than making them one-piece outputs.

Question 21.7 (Descriptive complexity). What are optimal deterministic or nondeterministic automaton sizes for these languages and their bounded-insertion stages? The arithmetic membership test is compact, but this alone is not a state bound. One can ask for a family with arbitrary prescribed plateaus and a target automaton whose state count is polynomial in the profile parameters.

Question 21.8 (Structural hypotheses restoring regular degree shapes). Which overlap restrictions on target shuffle sets force an interval spectrum, log-concavity, or real-rooted degree polynomials? Pairwise source separability eliminates the minimization conflict, but is much stronger than necessary. Bounds on the number of target representations per output, or laminarity of their low-cost regions, may offer intermediate conditions.

Question 21.9 (Infinite spectra over a fixed alphabet). Classify the spectra of infinite binary language pairs. A fixed inserted word a^m bounds all degrees by m , so an unbounded spectrum requires unbounded inserted length. Combining finite constructions across lengths is not automatic: cross-shuffles can introduce new degrees. A successful construction needs a mechanism controlling those interactions without a growing alphabet.

Question 21.10 (Formal verification and weighted extensions). A Lean development could start with finite vectors, the support formula, Lemmas 14.1 and 14.2, and the full-coverage lower bound. These are finite combinatorial statements and require no real analysis. A separate extension could assign different costs to insertion runs in different gaps, or track several inserted colors. Which finite sets of costs or vectors of costs admit analogous profile realizations?

22 Conclusion of Part II

The fixed-binary classification is as unrestricted as the earlier variable-alphabet classification: finiteness and the unavoidable degree 1 are the only conditions. Even one unary inserted word suffices. Its individual source pairs have identical, gap-free, real-rooted degree histograms, but minimization over targets can erase any chosen intermediate degrees and can prescribe every higher-degree multiplicity.

The gap-vector representation makes the mechanism explicit. Coordinatewise exclusion protects a designated expensive output; distance separation prevents other designated targets from reducing its cost; and a one-piece escape lemma makes all remaining outputs cheap. The same representation exposes a sharp extremal threshold: an isolated degree m in a full binary fixed-letter-count class needs m^2 symbols, and that threshold is attained for every $m \geq 6$. These are the proposed additions to the repository’s insertion-degree program.

23 A reproducible construction blueprint

Given $m \geq 2$ and a list $t_0, \dots, t_{q-1} \in \{2, \dots, m\}$, compute Λ , $\mathbf{b}_m$, $\mathbf{v}_i$, T , and β_i from Section 15. The following pseudocode is a direct implementation of the target definition.

```
def target_member(beta):
    if len(beta) != m or min(beta) < 0:
        return False
    if sum(beta) != T - m:
        return False
    if beta in special_targets:
        return True
    for v in centers:
        if all(beta[j] <= v[j] for j in range(m)):
            return False
    return True
```

Enumerating weak compositions of $T - m$ and applying this predicate lists all of B . Enumeration is not required to use the mathematical definition or the membership algorithm. To obtain the actual binary word from a vector, join the strings a^{β_j} with a single b between consecutive gaps. Zero gaps must be retained, not deleted from the vector.

24 Package contents, rebuilding and the choices of the merge

24.1 Shipped files

Manuscript 01 was delivered with its source `article.tex`, its compiled PDF, `code/verify.py`, the executed results in `data/`, a delivery README, `CLAIMS_AND_VALIDATION.md`, `SOURCES.md`, a Makefile and a checksum file. There are no external data downloads and no bundled font files. The report ships, beside Part I’s files:

Shipped path	Delivered as
<code>code/02-binary-profiles-verify.py</code>	<code>code/verify.py</code>
<code>code/02-binary-profiles-Makefile</code>	<code>Makefile</code>
<code>data/02-binary-profiles-verification.json</code>	<code>data/verification.json</code>
<code>data/02-binary-profiles-profile_checks.csv</code>	<code>data/profile_checks.csv</code>
<code>data/02-binary-profiles-worked_targets.csv</code>	<code>data/worked_targets.csv</code>
<code>data/02-binary-profiles-worked_outputs.csv</code>	<code>data/worked_outputs.csv</code>
<code>02-binary-profiles-CLAIMS_AND_VALIDATION.md</code>	<code>CLAIMS_AND_VALIDATION.md</code>
<code>02-binary-profiles-SOURCES.md</code>	<code>SOURCES.md</code>

The delivered article, PDF and README are printed or replaced by this report and its README; the checksum file was verified and not shipped. The shipped files are byte-identical to the delivered ones.

24.2 Rebuilding and rerunning

The article is rebuilt from the report directory with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Part II verifier needs Python 3.10 or later and only its standard library. *Its default output directory is the report's data/, where it would overwrite Part I's own data/verification.json and add unprefix CSV files.* Always pass an explicit output directory outside the report, for example

```
python code/02-binary-profiles-verify.py --out <scratch-directory>
```

and compare the results with the shipped data/02-binary-profiles-* files. The shipped code/02-binary-profiles-Makefile still uses the delivered layout: its `verify` target runs `code/verify.py`, which in this report is Part I's verifier, and its `all` and `clean` targets run `latexmk` on the merged `article.tex`. It is shipped as delivered and should not be used from the report directory.

24.3 Where the merge had to choose

1. Placement: the manuscript answers questions Part I names in its Section 10, so it became Part II of this report rather than a separate report; it is printed after Part I's Section 10 and before Part I's appendices, which contain no numbered items, so no number of Part I moved.
2. The definitions the manuscript repeats from Part I are printed once (Section 11.6); the manuscript's own bibliography entry for Part I is replaced by internal references, and its entry for Hughes's preprint is merged with Part I's (the same version, v1).
3. Twelve symbols were renamed where they clash with Part I (Table 1); no normalization changed.
4. The manuscript's title page, abstract and Section 1 became Sections 11.1–11.4; its two appendices became Sections 23 and 24, because the report's appendices belong to Part I.
5. Remark 11.1, the merge notes in Sections 19.3 and 20, the last paragraph after Corollary 12.2, and the dated pointers in Part I were added in the merge.

Part III. Sharp isolation thresholds for binary insertion

Part III was added on 30 September 2026 in batch 66 of ProveIt's incoming-report intake, from one manuscript written against Parts I and II as printed above. It answers Part II's Question 21.3 completely: the minimum lengths of an isolated full-coverage witness are 14, 21 and 27 for $m = 3, 4, 5$, the upper bounds Part II gave. It answers Question 21.4 for the centers and the mandatory targets, for every $m \geq 3$, and classifies the realizing target languages completely for $m = 3$. Parts I and II are unchanged except for dated pointers to this addition. Section 25 records the provenance and fixes the notation of the addition.

25 Provenance, conventions and the answer

25.1 One manuscript, one addition

Part III is batch-66 manuscript 04, *Sharp Isolation Thresholds for Binary Insertion: Exact parameter ranges, extremal centers, mandatory targets, and a complete degree-three classification*, an AI-assisted research manuscript dated 30 September 2026 (21 US-letter pages as delivered). Its title page reads “Research manuscript prepared with ChatGPT for Vladimir Reshetnikov” under the kicker “ProveIt research continuation”; that wording is recorded here as provenance only. It arrived as the archive `ProveIt_Sharp_Insertion_Isolation.zip` (inner directory of the same name) in ProveIt commit `9ad899cbe` and was placed in `4fee1cd07`. It was written against ProveIt commit `176e31c5c` (in full `176e31c5c7ee369edf51edd815463dc029e3faed`), at which this report's `article.tex` had the Git blob `7f5a958e` (in full `7f5a958e7ac6d6df902d4efa40cfefa813bb20db`) and its README the blob `bca8c085`. Both blobs are unchanged at the placement commit, so every statement the manuscript makes about “the repository report” refers to Parts I and II as printed above. Its “Theorem 12.3”, “Section 16” and “Questions 21.1, 21.3, 21.4 and 21.6” are Theorem 12.3, Section 16 and Questions 21.1, 21.3, 21.4 and 21.6, and are cited so below. The manuscript records that no repository file was changed in preparing it. Its code, data and audit files are shipped beside the files of Parts I and II with the prefix `03-sharp-isolation-`; its manuscript, delivery README and PDF are not shipped and survive in the arrival commit.

The manuscript contributes, in the order printed here: its answer and its statement of scope (this section); the gap representation and isolated centers (Section 26); the largest admissible target language (Section 27); the volume obstruction (Section 28); a complete criterion for an exceptional center (Section 29); all feasible parameters and the minimum lengths (Section 30); all extremal centers and their count (Section 31); the boundary family (Section 32); mandatory targets (Section 33); the complete degree-three classification (Section 34); residue classes and a counting bound (Section 35); its executed checks (Section 36); its proof audit and formalization boundary (Section 37); ten research questions (Section 38); and its conclusion (Section 39). Every result, proof, example, remark, table, limitation and question of the manuscript is printed. What it repeats from Parts I and II is printed once and cited (Section 25.8). The introductory note above, Sections 25.7, 25.8 and 40, and the dated pointers in Parts I and II were added in the merge.

25.2 The question of Part II

Part II's Theorem 12.3 proves a lower bound m^2 on the length of an isolated degree- m output in a full binary fixed-letter-count class, and constructs equality for every $m \geq 6$. Question 21.3 explicitly leaves open the exact minima for $m = 3, 4, 5$: the displayed lower bounds are 9, 16, 25, and the available constructions have

lengths 14, 21, 27. Question 21.4 asks which centers attain the sharp bound and which target words must occur in every realizing language.

Part III answers those questions in a precise sense. It determines all feasible centers and all feasible numerical parameters, not only the shortest examples. At the shortest length it gives an exact criterion for every center, an explicit test for every mandatory target, and a finite covering characterization of every realizing target language. For $m = 3$ that covering problem is solved completely, including the number of languages and their exact cardinality distribution. The general minimum number of target words is not claimed to be solved.

25.3 The answer in brief

Fix the inserted word a^m and require its shuffle with a finite binary language to fill a whole fixed-letter-count class, with exactly one output of insertion degree m and every other output of degree one. Part III gives a necessary and sufficient criterion for an arbitrary exceptional gap vector, including a boundary family outside Part II's strict escape criterion. It then determines every feasible parameter triple. Write g for the number of a -gaps, including the two end gaps, and T for the total number of a 's in an output. Thus the output contains $g - 1$ letters b and has length $N = T + g - 1$. Define

$$\tau_m = \begin{cases} 12, & m = 3, \\ 5m - 2, & m \geq 4. \end{cases} \quad (46)$$

Sections 30–34 prove the following statements.

For $m \geq 3$, a full-coverage isolated degree- m realization with parameters (g, T) exists exactly when

$$g \geq m, \quad T \geq \max\{g(m - 1) + 1, \tau_m\}.$$

Consequently the shortest possible length is

$$L_m = \begin{cases} 3, & m = 2, \\ 14, & m = 3, \\ 21, & m = 4, \\ 27, & m = 5, \\ m^2, & m \geq 6. \end{cases}$$

For $m = 3$ and $N = L_3$, there are exactly 19 683 target languages. Their minimum size is 37, attained by exactly 512 languages.

This settles the three small cases that Part II explicitly left open, while recovering its large- m bound. Part III also classifies all extremal centers, counts them by a finite binomial formula, and characterizes every mandatory target. At degree three, all shortest realizations reduce to 28 mandatory targets and nine disjoint two-element choices: there are 3^9 languages, of which 2^9 have minimum size 37. The proofs are self-contained apart from what is printed once in Part II; exact finite checks and reproducible code accompany them.

25.4 Why a new criterion is needed

Part II's construction protects an exceptional vector $\mathbf{v}$ by discarding all targets below it except a selected special target. A sufficient escape condition then makes every other output obtainable in one piece. A

sufficient condition alone cannot rule out shorter examples built from other centers, other target languages, or more gaps. The decisive step here is to replace that sufficient condition by a necessary and sufficient one.

There are two simplifications. First, there is a unique largest set of admissible targets preserving the exceptional degree. Any realization can be enlarged to that set. Second, a nonexceptional output with at least two eligible deletion coordinates always has a safe predecessor. The entire obstruction therefore lies in outputs with exactly one eligible coordinate. Their deficits are governed by coordinatewise lower bounds, which can be analyzed exactly.

25.5 Attribution and limits

The insertion convention follows Hughes [1, Sections 2 and 11], as in Parts I and II. His unary-source singleton interval theorem concerns one target word; allowing a target language introduces a minimization over different targets. The present argument does not resolve his conjecture for two arbitrary singleton words. The gap representation and the volume lower bound of Part II are restated below for completeness and explicitly credited to Part II. The new content is the exact obstruction analysis and its consequences, not the elementary observation that words can be represented by their run lengths.

The source search was targeted: the cited primary preprint, the pinned report and its questions, and searches on 30 September 2026 for the relevant insertion-degree terminology. No source found in that bounded check supplied the classification proved here. This is not an exhaustive priority search.

25.6 Status of the claims

Status (manuscript 04). These are written mathematical proofs in an unrefereed research manuscript, not Lean- or Rocq-certified theorems. The novelty claim is a specified advance over Part II as pinned, not a guarantee of historical priority. Full coverage, a unary inserted word, and a unique exception of maximum degree are essential hypotheses. Part III does not solve the unrestricted both-singleton conjecture, the general minimum target cardinality, or the shortest-witness problem without full coverage; its minimum target size 37 is proved at the shortest degree-three output length only. Nothing in Part III, as in Parts I and II, is formalized in any proof assistant, and no theorem inherits formal status from the Lean and Rocq developments beside which this collection sits. The finite computations of Section 36 are exact checks of stated ranges, and samples where so labelled, not substitutes for the proofs.

25.7 Notation of Part III

The symbols of Parts I and II keep their meanings. The manuscript's insertion, shuffle, degree and spectrum agree with Part I's, orientation included, and its gap vectors, fixed-letter-count classes and target sets with Part II's; Part III uses Part II's symbols for them. Where a symbol of the manuscript clashes with one of Parts I or II, Part III renames it; Table 2 lists the new symbols, the renamings, and the tempting false readings. No normalization was changed. Equations of Part III continue the report's single equation sequence, and theorem-like items and tables are numbered as in the rest of the report, so every number differs from the delivered PDF.

25.8 Material printed once

The manuscript restates, for completeness, several things already printed in Parts I and II. The definitions of insertion, shuffle and degree are Part I's (Section 2); they are recalled in Section 26 with the manuscript's

Table 2: Symbols of Part III. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

Symbol	Meaning in Part III	Not to be confused with
$\mathcal{I}_k(A, B)$ (renamed from $I_k(A, B)$)	Part I’s k -insertion (1)	—
$C_{T,g}, \mathcal{W}_{T,g}, \omega(\mathbf{c})$ (renamed from $C(T, g), \mathcal{W}(T, g), \omega(\mathbf{c})$)	Part II’s weak compositions, binary fixed-letter-count class and gap-vector word	Part I’s typewriter words w
$\mathcal{E}, B_{\mathcal{E}}, d_{\mathcal{E}}(\mathbf{c})$ (renamed from E, B_E, d_E)	a set of target gap vectors (as in Part II), its target language, and the degree of $\omega(\mathbf{c})$ against it	Part I’s E , the bad adjacent pairs of (9)
eligible; $P_m(\mathbf{c})$	$c_j \geq m$; the one-piece predecessors (51)	Part I’s grammar nonterminals P_j ; the pairs $\mathcal{P}_i$ of Section 34
$P_m^{\text{adm}}(\mathbf{c})$ (renamed from $Q(\mathbf{c})$)	the admissible predecessors of Section 33	$Q = \Sigma $ and $Q_{w,k}$ of Part I
$\mathcal{A}_m(\mathbf{v}), \mathcal{S}_m(\mathbf{v})$	admissible and special targets, (53)–(54)	the spectrum $\mathcal{D}(A, B)$; the inserted language A
h_i, H	excess heights and their sum, (59)	Part II’s counts $h_{m,g,r}$ and polynomial $H_m(z)$
$\hat{h}$ (renamed from M)	the maximum height	Part I’s M in its search encoding; Part II’s μ ; the mandatory set $\mathcal{M}$
ρ_j (renamed from H_j)	$H - h_j$: the same quantity as Part II’s escape sum $\rho_j(\mathbf{v})$	—
$\tau_m, T_{m,g}, T_m$	the mass threshold (46), the least feasible total with g gaps (69), the total at the shortest length (73)	the letter count T
$L_m, L_{m,g}$	minimum output lengths	a language
e_m	number of ordered extremal centers (77)	the unit vectors $\mathbf{e}_j$
$\text{Comp}(n, r, b)$ (renamed from $C(n, r, b)$)	bounded positive compositions (76)	$C_{T,g}$
$\text{Hi}(\beta)$ (renamed from $K(\beta)$)	the high coordinates $\{i : \beta_i \geq m\}$ of a target	Part I’s degree bound K ; Part II’s κ_m
ν (renamed from K)	quotient total of a residue class, Proposition 35.1	—
k	number of high coordinates (Lemma 30.1)	the insertion bound k of $\mathcal{I}_k(A, B)$
$\mathcal{A}, \mathcal{M}, \mathcal{P}_1, \dots, \mathcal{P}_9$	at $m = 3$: $\mathcal{A}_m(\mathbf{v})$ for $\mathbf{v} = (4, 4, 4)$, the mandatory targets, the nine pairs	the inserted language A ; $P_m(\mathbf{c})$
$L_m = m^2$ ($m \geq 6$)	minimum length in the full-coverage class	not a lower bound for every word of degree m : that bound is $2m - 1$

indexing. Its gap-degree lemma (Lemma 26.1) is Part II's Lemma 13.1 with Corollary 13.2; its statement is kept and its proof, the same argument, is printed once, in Part II. Its volume bound (Proposition 28.1) is Part II's Proposition 16.2 plus a support bound; the manuscript's proof is kept because it reaches $g \geq m$ by a different route. Its minimum length m^2 for $m \geq 6$ is Part II's Theorem 12.3, as the manuscript says. The maximal degree-three target set (80) is the target set of Part II's worked example (39), and Part II's optimal center (36) is one of the extremal centers classified in Theorem 31.1. The manuscript's bibliography entry for the pinned report is replaced by these internal references, and its entry for Hughes's preprint by Part I's (the same version, v1).

26 Insertion and finite gap geometry

26.1 Conventions

For languages A, B over a finite alphabet and $k \geq 1$, recall Part I's definition (1), here with the target factors indexed from 1:

$$\mathcal{I}_k(A, B) = \{x_1 y_1 x_2 y_2 \cdots x_k y_k x_{k+1} : y_1 \cdots y_k \in A, x_1 \cdots x_{k+1} \in B\}. \quad (47)$$

Empty factors are allowed. The inserted source is A , not B . The shuffle $A \sqcup B$ is the union of these sets. For a shuffle output w , its insertion degree $d_{A,B}(w)$ is the least k for which $w \in \mathcal{I}_k(A, B)$. Equivalently, it is the minimum number of runs of inserted-source positions among all masks producing w (Lemma 2.2).

Throughout the principal theorems, $A = \{a^m\}$, the alphabet is exactly $\{a, b\}$, and $m \geq 3$. The small case $m = 2$ is treated separately. For $T \geq 0$ and $g \geq 1$, let, as in Part II (Section 12),

$$C_{T,g} = \{(c_1, \dots, c_g) \in \mathbb{N}_0^g : \sum_i c_i = T\}.$$

The word associated to $\mathbf{c} \in C_{T,g}$ is

$$\omega(\mathbf{c}) = a^{c_1} b a^{c_2} b \cdots b a^{c_g}. \quad (48)$$

This includes zero-length gaps and therefore consecutive b 's and words beginning or ending with b . The correspondence is a bijection onto the fixed-letter-count class $\mathcal{W}_{T,g}$, which has cardinality $\binom{T+g-1}{g-1}$. A target language of the required counts has the form

$$B_{\mathcal{E}} = \{\omega(\boldsymbol{\beta}) : \boldsymbol{\beta} \in \mathcal{E}\}, \quad \mathcal{E} \subseteq C_{T-m,g}.$$

This restriction on target counts is forced by full coverage with no extra outputs: inserting a^m adds exactly m copies of a and leaves the number of b 's unchanged. Coordinatewise comparison is denoted by $\boldsymbol{\beta} \leq \mathbf{c}$. We abbreviate $d_{\mathcal{E}}(\mathbf{c}) = d_{\{a^m\}, B_{\mathcal{E}}}(\omega(\mathbf{c}))$.

Lemma 26.1 (Gap degree; Part II background). *For a fixed target $\boldsymbol{\beta}$, an output $\mathbf{c}$ is obtainable by inserting a^m exactly when $\boldsymbol{\beta} \leq \mathbf{c}$ and $\sum_i (c_i - \beta_i) = m$. When these conditions hold,*

$$d_{\{a^m\}, \{\omega(\boldsymbol{\beta})\}}(\omega(\mathbf{c})) = |\text{supp}(\mathbf{c} - \boldsymbol{\beta})|. \quad (49)$$

Thus, for a target set $\mathcal{E}$,

$$d_{\mathcal{E}}(\mathbf{c}) = \min_{\substack{\boldsymbol{\beta} \in \mathcal{E} \\ \boldsymbol{\beta} \leq \mathbf{c}}} |\text{supp}(\mathbf{c} - \boldsymbol{\beta})|, \quad (50)$$

where an empty minimum means that $\omega(\mathbf{c})$ is not a shuffle output.

Proof. This is Lemma 13.1 with Corollary 13.2, whose proofs are printed once, in Part II. The manuscript reproves them by the same argument: all b 's come from the target, a b separates each receiving gap from every other, and putting the new letters of each receiving gap consecutively attains the bound. Like Part II's proof, it concerns the minimum over masks, not uniqueness of a position mask. $\square$

A coordinate j of $\mathbf{c}$ is *eligible* when $c_j \geq m$. Its one-piece predecessor is $\mathbf{c} - m\mathbf{e}_j$. Define

$$P_m(\mathbf{c}) = \{\mathbf{c} - m\mathbf{e}_j : c_j \geq m\}. \quad (51)$$

By Lemma 26.1, $d_{\mathcal{E}}(\mathbf{c}) = 1$ exactly when $P_m(\mathbf{c}) \cap \mathcal{E}$ is nonempty; this is Part II's criterion (17).

Definition 26.2 (Isolated maximum-degree center). A vector $\mathbf{v} \in C_{T,g}$ is *feasible at inserted length m* when some $\mathcal{E} \subseteq C_{T-m,g}$ satisfies

$$\{a^m\} \sqcup B_{\mathcal{E}} = \mathcal{W}_{T,g}, \quad d_{\mathcal{E}}(\mathbf{v}) = m, \quad d_{\mathcal{E}}(\mathbf{c}) = 1 \quad (\mathbf{c} \neq \mathbf{v}). \quad (52)$$

Such an $\mathcal{E}$ is a realizing target set. The word $\omega(\mathbf{v})$ is the unique exceptional output.

The requirement is stronger than having spectrum $\{1, m\}$: it includes full coverage and multiplicity one for the high degree. Those two requirements are used in the lower bounds.

27 The largest admissible target language

Definition 27.1 (Admissible and special targets). For $\mathbf{v} \in C_{T,g}$, set

$$\mathcal{S}_m(\mathbf{v}) = \{\mathbf{v} - \mathbf{1}_S : S \subseteq \text{supp}(\mathbf{v}), |S| = m\}, \quad (53)$$

$$\mathcal{A}_m(\mathbf{v}) = \{\boldsymbol{\beta} \in C_{T-m,g} : \boldsymbol{\beta} \not\leq \mathbf{v}\} \cup \mathcal{S}_m(\mathbf{v}). \quad (54)$$

Here $\mathbf{1}_S$ is the 0–1 indicator vector of S . Targets in the first set of (54) are called *safe*.

Lemma 27.2 (Maximal admissible language). *Suppose $\mathbf{v}$ is feasible. Then $|\text{supp}(\mathbf{v})| \geq m$, every realizing set $\mathcal{E}$ is contained in $\mathcal{A}_m(\mathbf{v})$, and $\mathcal{E} \cap \mathcal{S}_m(\mathbf{v}) \neq \emptyset$. Conversely, if the special set is nonempty, then $\mathcal{A}_m(\mathbf{v})$ gives degree m to $\mathbf{v}$. Consequently $\mathbf{v}$ is feasible if and only if*

$$|\text{supp}(\mathbf{v})| \geq m \quad \text{and} \quad P_m(\mathbf{c}) \cap \mathcal{A}_m(\mathbf{v}) \neq \emptyset \quad \text{for every } \mathbf{c} \in C_{T,g} \setminus \{\mathbf{v}\}. \quad (55)$$

Proof. If $\boldsymbol{\beta} \leq \mathbf{v}$ is a realizing target, the deficit $\mathbf{v} - \boldsymbol{\beta}$ has total m . It has support m precisely when its m positive coordinates are all one. Any other deficit has support less than m and would lower the exceptional degree. Thus every accepted target below $\mathbf{v}$ must be special, and at least one must exist. All targets not below $\mathbf{v}$ are harmless to its degree. This proves the containment, and also shows that enlarging a realization to $\mathcal{A}_m(\mathbf{v})$ preserves the exceptional degree while retaining every one-piece representation. The reverse implication follows directly from Lemma 26.1. $\square$

Corollary 27.3 (Exact characterization of realizing languages). *For any feasible center, a target set $\mathcal{E}$ realizes it exactly when*

$$\mathcal{E} \subseteq \mathcal{A}_m(\mathbf{v}), \quad \mathcal{E} \cap \mathcal{S}_m(\mathbf{v}) \neq \emptyset, \quad \mathcal{E} \cap P_m(\mathbf{c}) \neq \emptyset \quad (\mathbf{c} \neq \mathbf{v}). \quad (56)$$

In particular, when $g = m$, every coordinate of a feasible center is positive and its unique special target is

$$\boldsymbol{\beta}_* = \mathbf{v} - \mathbf{1}. \quad (57)$$

This reduction loses no target languages: it is not an assumption that a realization has the particular safe-plus-one form used in Part II's construction (Theorem 14.4). It also makes explicit why enlarging the language is useful for existence but not for minimizing its cardinality.

28 The unavoidable volume obstruction

The next proposition is the lower-bound argument of Part II (Section 16: Lemma 16.1 and Proposition 16.2), included with its boundary case because the strict inequality is needed later. Its bounds on g , T and N are Proposition 16.2; the support bound is added here.

Proposition 28.1 (Support and volume bounds). *If $m \geq 3$ and $\mathbf{v} \in C_{T,g}$ is feasible, then*

$$g \geq m, \quad |\text{supp}(\mathbf{v})| \geq m, \quad T > g(m-1). \quad (58)$$

In particular $N = T + g - 1 \geq gm \geq m^2$.

The manuscript's proof follows Part II's proof of Proposition 16.2, but obtains $g \geq m$ from the support bound; it is kept as a second route.

Second proof (manuscript 04). The support assertion follows from Lemma 27.2; it implies $g \geq m$ and $T \geq m$. A vector whose coordinates are all at most $m-1$ has no one-piece predecessor, so full coverage makes it an exceptional output. If $0 < T < g(m-1)$, there are at least two such vectors. For example, fill coordinates successively to $m-1$ and then use any remainder. At an interior total the resulting vector is not constant, so a permutation gives a second vector of the same total. This contradicts uniqueness.

If $T = g(m-1)$, the capped vector is uniquely $\mathbf{v}_* = (m-1, \dots, m-1)$, so it must be the exceptional center. Its neighbor

$$\mathbf{c} = (m, m-2, m-1, \dots, m-1)$$

is distinct and has only its first coordinate eligible. Its unique one-piece predecessor

$$\boldsymbol{\beta} = (0, m-2, m-1, \dots, m-1)$$

must therefore be accepted. But $\mathbf{v}_* - \boldsymbol{\beta} = (m-1, 1, 0, \dots, 0)$ has support $2 < m$, a contradiction. Hence the total must be strictly greater than the cap, proving (58) and the length inequalities. $\square$

The same neighbor has support two when $m = 2$, so it does not contradict a degree-two exception. This explains, rather than hides, the restriction $m \geq 3$ in the principal classification.

29 A complete criterion for an exceptional center

29.1 Statement

For a vector $\mathbf{v}$ define its excess heights above the ineligible cap by

$$h_i = (v_i - m + 1)_+, \quad H = \sum_i h_i, \quad \hat{h} = \max_i h_i, \quad \rho_j = H - h_j. \quad (59)$$

A coordinate is called high if $h_i > 0$, equivalently $v_i \geq m$. The sum $\rho_j = \sum_{i \neq j} h_i$ is Part II's escape sum $\rho_j(\mathbf{v})$ of (20); the manuscript writes it H_j , and its maximum height $\hat{h}$ is the manuscript's M .

Theorem 29.1 (Complete center classification). *Let $m \geq 3$, $g \geq 1$, and $\mathbf{v} \in C_{T,g}$. Then $\mathbf{v}$ is feasible if and only if $|\text{supp}(\mathbf{v})| \geq m$, $T > g(m-1)$, and one of the following mutually exclusive conditions holds:*

$$H - \hat{h} \geq m + 1; \quad \text{the strict case,} \quad (60)$$

$$H - \hat{h} = m \text{ and, after removing one maximum } h_j = \hat{h}, \text{ all remaining } h_i \leq 1; \quad \text{the boundary case.} \quad (61)$$

In the boundary case the positive height multiset is exactly

$$\{\underbrace{\hat{h}, 1, \dots, 1}_{m \text{ copies}}, \quad \hat{h} \geq 1. \quad (62)$$

Equivalently, in place of (60)–(61), one may require for every coordinate j that

$$\rho_j > m \quad \text{or} \quad (\rho_j = m \text{ and } h_i \leq 1 \text{ for every } i \neq j). \quad (63)$$

Whenever the criterion holds, the maximal target set $\mathcal{A}_m(\mathbf{v})$ realizes $\mathbf{v}$.

Part II's escape lemma (Lemma 14.2, whose hypothesis $\rho_j(\mathbf{v}) > m$ for all j says $H - \hat{h} \geq m + 1$) gives the strict case as a sufficient condition. The necessity statement, the exact boundary alternative, and the reduction to arbitrary target languages are the additional content here.

29.2 Reduction to a unique eligible coordinate

Lemma 29.2 (At most one unsafe predecessor). *For $\mathbf{c} \neq \mathbf{v}$ of the same total, at most one member of $P_m(\mathbf{c})$ is coordinatewise below $\mathbf{v}$. In particular, if $\mathbf{c}$ has at least two eligible coordinates, it has a safe one-piece predecessor.*

Proof. Because $\mathbf{c}$ and $\mathbf{v}$ have equal totals and are distinct, some coordinate i satisfies $c_i > v_i$. If $\mathbf{c} - m\mathbf{e}_j \leq \mathbf{v}$, every excess coordinate must equal j , since all other coordinates are unchanged. Thus two distinct deletion coordinates cannot both yield predecessors below $\mathbf{v}$. For the last statement, choose an eligible coordinate different from an excess coordinate. The excess survives that deletion. $\square$

The volume inequality supplies at least one eligible coordinate for every output. The only possible failure is therefore a vector $\mathbf{c} \neq \mathbf{v}$ with one eligible coordinate j , whose sole predecessor β is forbidden. Writing $\delta = \mathbf{v} - \beta$, this failure has the exact description

$$\begin{aligned} 0 \leq \delta \leq \mathbf{v}, \quad \sum_i \delta_i = m, \quad \delta_i \geq h_i \quad (i \neq j), \\ \delta \neq m\mathbf{e}_j, \quad \delta \notin \{0, 1\}^g. \end{aligned} \quad (64)$$

Indeed, $\mathbf{c} = \mathbf{v} - \delta + m\mathbf{e}_j$, its non- j coordinates are below m exactly when the displayed lower bounds hold, and $\mathbf{c} \neq \mathbf{v}$ exactly when $\delta \neq m\mathbf{e}_j$. The final condition says that the predecessor is not special. Conversely, any deficit satisfying (64) constructs a nonexceptional output with a unique forbidden predecessor. Thus it is an explicit impossibility certificate.

29.3 Necessity of the local inequalities

Lemma 29.3 (At least two high coordinates). *A feasible center for $m \geq 3$ has at least two high coordinates.*

Proof. There is at least one by the volume bound. If j were the only one, choose a different positive coordinate i , which exists because the support has size at least m . Set $\delta = (m-1)\mathbf{e}_j + \mathbf{e}_i$. It is bounded by $\mathbf{v}$, has total m , and is not binary. Every non- j coordinate of $\mathbf{v}$ is at most $m-1$, so (64) holds. This contradicts feasibility. $\square$

Lemma 29.4 (Exact unique-predecessor obstruction). *Assume $m \geq 3$, $T > g(m-1)$, support size at least m , and at least two high coordinates. There is no obstruction (64) at coordinate j exactly when the local condition (63) holds at j .*

Proof. At least one high coordinate lies outside j . Thus every deficit satisfying the outside lower bounds has a positive outside entry and cannot equal $m\mathbf{e}_j$.

If $\rho_j > m$, the lower bounds alone exceed the required deficit total, so no such deficit exists. If $\rho_j = m$, the only possible deficit has $\delta_j = 0$ and $\delta_i = h_i$ for $i \neq j$. It is binary exactly when all those outside heights are at most one. This proves the assertion at and above the boundary.

Suppose $\rho_j < m$. Begin with the vector having outside entries h_i and entry zero at j . If an outside height is at least two, the initial vector already has a nonbinary entry. Otherwise choose an outside high coordinate i ; its height is one and its capacity v_i is at least $m \geq 3$. Increase that entry from one to two. In both cases the current sum is at most m , all entries are within their capacities, the outside lower bounds hold, and a nonbinary entry has been created. Complete the vector to total m by adding units without exceeding $\mathbf{v}$. This is possible because the sum of all capacities is $T \geq m$: whenever the current sum is below m , not all capacities have been filled. The nonbinary entry remains, and (64) follows. Thus $\rho_j < m$ always creates an obstruction. $\square$

29.4 Proof of the classification theorem

Proof of Theorem 29.1. Necessity of the support and volume bounds is Proposition 28.1. Necessity of at least two high coordinates is Lemma 29.3. Using Lemmas 27.2, 29.2 and 29.4, feasibility is now equivalent to the local inequalities (63).

Their minimum left side is $H - \hat{h}$. If $H - \hat{h} > m$, every local inequality is strict. If $H - \hat{h} < m$, the inequality fails at a maximum coordinate. If $H - \hat{h} = m$, applying the local condition at a maximum coordinate forces every outside height to be zero or one, with exactly m ones. Conversely, suppose that pattern holds. For a coordinate of height less than $\hat{h}$, its outside sum is strictly greater than m . At any maximum coordinate the outside heights are all at most one: if $\hat{h} > 1$ the maximum is unique, and if $\hat{h} = 1$ every positive height is one. Thus the local inequalities hold everywhere.

For sufficiency, the two alternatives themselves supply at least two high coordinates. Each nonexceptional output has an eligible coordinate by the volume bound. If it has at least two, Lemma 29.2 supplies a safe predecessor. If it has only one and that predecessor lies below $\mathbf{v}$, the local condition forces a binary deficit, so that predecessor is special and admissible. The special set is nonempty by the support assumption. Therefore Lemma 27.2 proves that $\mathcal{A}_m(\mathbf{v})$ realizes the center. $\square$

29.5 An effective criterion and negative certificates

Computing the support size, T , H , $\hat{h}$, and the condition outside a maximum takes $O(g)$ arithmetic operations and comparisons. This is an operation count on integer inputs, not an assertion of constant bit cost for arbitrary-size integers. It avoids enumerating $\binom{T+g-1}{g-1}$ outputs or all target languages.

If the support or volume test fails, the proof gives a support obstruction or two capped outputs, with the explicit boundary neighbor at equality. If those tests pass but the height test fails, the proof of Lemma 29.3 or Lemma 29.4 constructs a deficit and hence an output with a unique forbidden predecessor. A checker can validate that negative certificate by additions, inequalities, and support counting. The accompanying code includes both the linear classifier and independent finite obstruction enumerators.

30 All feasible numerical parameters

30.1 A total-mass obstruction beyond the volume bound

The volume bound grows with g , but does not explain the three small exceptions to m^2 . A second lower bound comes from the number of high coordinates and the excess mass they must carry.

Lemma 30.1 (Minimum total mass). *Every feasible center at inserted length $m \geq 3$ satisfies*

$$T \geq \tau_m, \quad (65)$$

with τ_m as in (46).

Proof. Consider first the boundary case. Its $m + 1$ positive heights require at least $m + 1$ coordinates of $\mathbf{v}$ to be at least m . Hence $T \geq m(m + 1)$. For $m = 3$ this equals $\tau_3 = 12$; for $m \geq 4$ it is strictly greater than $5m - 2$, since $m(m + 1) - (5m - 2) = (m - 2)^2 - 2 > 0$.

In the strict case let k be the number of high coordinates. Then $k \geq 2$, $H - \hat{h} \geq m + 1$, and $H \leq k\hat{h}$. Therefore

$$\hat{h} \geq \left\lceil \frac{m + 1}{k - 1} \right\rceil, \quad H \geq m + 1 + \left\lceil \frac{m + 1}{k - 1} \right\rceil. \quad (66)$$

The high coordinates contribute $k(m - 1) + H$ to T . If $k < m$, at least $m - k$ additional coordinates must be positive to provide support size m . Thus

$$T \geq k(m - 1) + H + (m - k)_+. \quad (67)$$

If $k > m$, the simpler estimate $T \geq km \geq m(m + 1)$ suffices. For $2 \leq k \leq m$, combine (66) and (67) to get

$$T \geq 2m + 1 + f_m(k), \quad f_m(k) = k(m - 2) + \left\lceil \frac{m + 1}{k - 1} \right\rceil. \quad (68)$$

When $m = 3$, the possibilities are $f_3(2) = 6$ and $f_3(3) = 5$, giving $T \geq 12$. For $m \geq 4$, $f_m(2) = 3m - 3$. At $k = 3$,

$$f_m(3) = 3m - 6 + \left\lceil \frac{m + 1}{2} \right\rceil \geq 3m - 3,$$

with equality exactly for $m = 4, 5$. For $4 \leq k \leq m$ the ceiling is at least two, and therefore

$$f_m(k) \geq 4(m - 2) + 2 = 4m - 6 > 3m - 3.$$

Consequently $T \geq 2m + 1 + 3m - 3 = 5m - 2$ for $m \geq 4$. □

30.2 Sharp threshold with a prescribed number of gaps

Theorem 30.2 (Exact parameter triples). *Let $m \geq 3$, $g \geq 1$, and $T \geq 0$. There is a feasible center in $C_{T,g}$ if and only if*

$$g \geq m, \quad T \geq T_{m,g} := \max\{g(m - 1) + 1, \tau_m\}. \quad (69)$$

For every permitted triple there is an explicit realization with exactly m positive center coordinates and hence a unique special target.

Proof. Necessity is Proposition 28.1 and Lemma 30.1. For sufficiency start with a vector having g coordinates:

$$\mathbf{v}^{(0)} = \begin{cases} (4, 4, 4, 0, \dots, 0), & m = 3, \\ (2m, 2m, \underbrace{1, \dots, 1}_{m-2}, 0, \dots, 0), & m \geq 4. \end{cases} \quad (70)$$

Its total is τ_m and its support size is m . For $m = 3$ the positive heights are $(2, 2, 2)$, so $H - \hat{h} = 4 = m + 1$. For $m \geq 4$ they are $(m + 1, m + 1)$, again with $H - \hat{h} = m + 1$. Add $T - \tau_m \geq 0$ to the first coordinate. That coordinate was already one of the maxima, so its height and the total height increase by the same amount; $H - \hat{h}$ stays $m + 1$. The support remains m , and the requested bound on T makes the volume inequality strict. The strict branch of Theorem 29.1 applies. Use the target language $\mathcal{A}_m(\mathbf{v})$, whose only special target subtracts one at each positive coordinate of $\mathbf{v}$. $\square$

For fixed m and $g \geq m$, the shortest output length is therefore

$$L_{m,g} = T_{m,g} + g - 1 = \max\{gm, \tau_m + g - 1\}. \quad (71)$$

This formula resolves the possibility, raised in Question 21.3, that increasing the number of gaps might shorten a small witness. Both terms are strictly increasing in g , so it cannot.

Theorem 30.3 (Exact minimum length). *Among all full-coverage isolated degree- m realizations with inserted word a^m , the minimum output length is*

$$L_m = \begin{cases} 3, & m = 2, \\ 14, & m = 3, \\ 21, & m = 4, \\ 27, & m = 5, \\ m^2, & m \geq 6. \end{cases} \quad (72)$$

For every $m \geq 3$, a shortest realization necessarily has $g = m$.

Proof. For $m \geq 3$, minimize (71) over $g \geq m$. Strict monotonicity forces $g = m$, giving

$$L_m = \max\{m^2, \tau_m + m - 1\}.$$

The displayed small values follow directly. For $m \geq 6$, the second term is $6m - 3$, and $m^2 \geq 6m - 3$ because $m^2 - 6m + 3$ is positive at $m = 6$ and increases thereafter. The construction in Theorem 30.2 attains these values.

For $m = 2$, any degree-two mask needs two inserted letters separated by a target letter, so its length is at least three. The language $A = \{aa\}$, $B = \{b\}$ has outputs aab, aba, baa , with respective degrees 1, 2, 1. It fills $\mathcal{W}_{2,2}$ and attains the bound. $\square$

The $m \geq 6$ value was already proved in Part II (Theorem 12.3). The new length assertions are optimality of the three small examples and the complete fixed-gap formula, supported by the stronger necessity theorem.

31 All extremal centers and their enumeration

31.1 Classification at the shortest length

Let

$$T_m = L_m - m + 1 \quad (m \geq 3). \quad (73)$$

Theorem 31.1 (Extremal center classification). *The centers of shortest realizations are exactly the positive integer m -tuples $\mathbf{v}$ with total T_m satisfying*

$$\sum_{i=1}^m (v_i - m + 1)_+ - \max_{1 \leq i \leq m} (v_i - m + 1)_+ \geq m + 1. \quad (74)$$

Up to coordinate permutation, the small cases are precisely

m	center types
3	(4, 4, 4),
4	(8, 8, 1, 1), (6, 6, 5, 1),
5	(10, 10, 1, 1, 1), (7, 7, 7, 1, 1).

(75)

For $m \geq 6$, the criterion together with $\sum_i v_i = m^2 - m + 1$ is a complete parametric classification; it is not restricted to Part II's two-large-coordinate family (36).

Proof. A shortest realization has $g = m$ by Theorem 30.3. Its support must therefore be all coordinates. The boundary branch requires at least $m + 1$ high coordinates and is impossible. The full criterion reduces exactly to (74); the volume bound is satisfied by T_m .

For $m = 3$, equality $T = 12$ in the mass argument forces $k = 3$, $H = 6$, $\hat{h} = 2$, so all three heights are two and the center is (4, 4, 4). For $m = 4$, equality $T = 18$ permits only $k = 2$ or $k = 3$. In the first case all low positive coordinates equal one, $H = 10$, and $\hat{h} = 5$, giving high heights (5, 5) and center (8, 8, 1, 1). In the second case the low coordinate is one, $H = 8$, and $\hat{h} = 3$; the positive heights must be (3, 3, 2), giving (6, 6, 5, 1).

For $m = 5$ and $T = 23$, the analogous $k = 2$ case has heights (6, 6) and three low ones, giving (10, 10, 1, 1, 1). The $k = 3$ case has $H = 9$, $\hat{h} = 3$, so the heights are (3, 3, 3) and the low coordinates are both one. This gives (7, 7, 7, 1, 1). The strict inequalities for all other k in Lemma 30.1 exclude further types. $\square$

Coordinate permutation changes the order of gaps and in general changes the binary word. It preserves feasibility, so a list “up to permutation” is an orbit list, not a count of words. In particular the two types at $m = 4$ yield $6 + 12 = 18$ distinct centers, and those at $m = 5$ yield $10 + 10 = 20$.

31.2 A finite formula for the number of centers

Let $\text{Comp}(n, r, b)$ count positive compositions of n into r ordered parts, each at most b . Set $\text{Comp}(n, 0, b) = 1$ if $n = 0$ and zero otherwise. For $r > 0$ set it to zero if $n < r$, $b < 1$, or $n > rb$. In all other cases,

$$\text{Comp}(n, r, b) = \sum_{j=0}^{\min\{r, \lfloor (n-r)/b \rfloor\}} (-1)^j \binom{r}{j} \binom{n-jb-1}{r-1}. \quad (76)$$

This follows by subtracting one from each coordinate and applying inclusion–exclusion to the coordinates exceeding the resulting cap $b - 1$.

Theorem 31.2 (Exact extremal-center count). *For $m \geq 3$, let e_m be the number of distinct ordered gap vectors of shortest realizations. Then*

$$e_m = \sum_{k=2}^m \binom{m}{k} \sum_{H=k}^{T_m} \text{Comp}(H, k, H - m - 1) \text{Comp}(T_m - k(m - 1) - H, m - k, m - 1). \quad (77)$$

This counts centers, not target languages, source masks, or permutation orbits.

Proof. Choose the k high positions in $\binom{m}{k}$ ways. Their positive heights sum to H . The strict condition $H - \hat{h} \geq m + 1$ is exactly the condition that each such height is at most $H - m - 1$, giving the first composition factor. The corresponding center coordinates contribute $k(m - 1) + H$. Every remaining center coordinate lies in $\{1, \dots, m - 1\}$ and their sum is $T_m - k(m - 1) - H$, giving the second factor. Every extremal center has one unique choice of these data. The zero conventions discard all impossible parameter values. $\square$

Table 3: Exact ordered-center counts from (77). Orbit enumeration independently checks the formula through $m = 7$.

m	L_m	e_m	permutation orbits, when enumerated
3	14	1	1
4	21	18	2
5	27	20	2
6	36	2 220	19
7	49	677 985	660
8	64	79 327 528	—
9	81	6 759 129 996	—
10	100	520 406 950 830	—
11	121	39 849 098 160 190	—
12	144	3 174 502 201 343 220	—

For example, $m = 6$ already has 19 unordered types, rather than one. The shipped file `data/03-sharp-isolation-extremal_orbits_m3_m6.json` records all of them, together with the small-case lists. The first few include $(1, 1, 1, 1, 12, 15)$, $(1, 1, 1, 1, 13, 14)$, $(1, 1, 1, 6, 11, 11)$, and $(1, 1, 1, 9, 9, 10)$. The count formula is a theorem; the displayed integers are exact evaluations of it, not floating-point estimates.

32 The boundary family and its sharp cost

The extra branch in Theorem 29.1 is not an empty exception. It captures centers for which some unique-predecessor deficits are forced to be binary and are therefore harmless.

Proposition 32.1 (Shortest non-strict centers). *A feasible center outside the strict branch has*

$$g \geq m + 1, \quad T \geq m(m + 1), \quad N \geq m^2 + 2m. \quad (78)$$

These bounds are simultaneously attained by

$$g = m + 1, \quad \mathbf{v} = (m, m, \dots, m). \quad (79)$$

Its $m + 1$ special targets are obtained by subtracting one from all but one coordinate.

Proof. The boundary multiset (62) has $m + 1$ positive heights, so there are at least $m + 1$ high coordinates and each contributes at least m to the total. This proves the first two inequalities and hence the length bound. For (79), every height is one, $H - \hat{h} = m$, the support is $m + 1$, and $T = m(m + 1) > (m + 1)(m - 1)$. The boundary branch applies. The description of the special targets is (53). $\square$

For $m = 3$ this gives center $(3, 3, 3, 3)$ at total 12 and length 15. The output $(6, 2, 2, 2)$ has only one eligible coordinate. Its sole predecessor $(3, 2, 2, 2)$ is below the center, so it is not safe; nevertheless its deficit $(0, 1, 1, 1)$ is binary and the predecessor is special. This is exactly why demanding strict escape would reject a valid center.

The boundary examples do not improve the global optimum. They instead show that a proof of necessity based on Part II's strict escape condition alone would be false. Their sharp cost (78) explains why they do not appear in the shortest-center classification.

33 Mandatory targets in every realizing language

33.1 The exact covering criterion

For a feasible center write

$$P_m^{\text{adm}}(\mathbf{c}) = P_m(\mathbf{c}) \cap \mathcal{A}_m(\mathbf{v}) \quad (\mathbf{c} \neq \mathbf{v}).$$

Every $P_m^{\text{adm}}(\mathbf{c})$ is nonempty. A target is *mandatory* if it belongs to every realizing target set for that center. The following elementary covering observation is useful because the ambient maximal language is already known explicitly.

Proposition 33.1 (Mandatory targets, general form). *A target $\beta \in \mathcal{A}_m(\mathbf{v})$ is mandatory if and only if either $S_m(\mathbf{v}) = \{\beta\}$ or $P_m^{\text{adm}}(\mathbf{c}) = \{\beta\}$ for some $\mathbf{c} \neq \mathbf{v}$.*

Proof. A singleton special set or singleton predecessor constraint forces its only member by Corollary 27.3. Conversely, remove β from $\mathcal{A}_m(\mathbf{v})$. If neither singleton condition holds, at least one special target remains and every nonexceptional output retains an admissible predecessor. Thus the reduced language still realizes the center, proving that β was not mandatory. $\square$

The proposition concerns removal of one target at a time. Several individually optional targets may not be simultaneously removable: they may be the only alternatives for a common output.

33.2 A closed test at the sharp length

For a target β set

$$\text{Hi}(\beta) = \{i : \beta_i \geq m\}.$$

At any extremal center $g = m$ and the special target is $\beta_* = \mathbf{v} - \mathbf{1}$.

Theorem 33.2 (Explicit mandatory-target classification). *Let $m \geq 3$, $g = m$, and let $\mathbf{v}$ be any feasible center, not necessarily of minimum total. For $\beta \in \mathcal{A}_m(\mathbf{v})$, it is mandatory exactly in the following cases:*

- (i) $\beta = \beta_*$;
- (ii) $|\text{Hi}(\beta)| \leq 1$;

- (iii) $\text{Hi}(\beta) = \{p, q\}$ and at least one of $\beta + m\mathbf{e}_p - m\mathbf{e}_q$ or $\beta + m\mathbf{e}_q - m\mathbf{e}_p$ is forbidden, that is, is not in $\mathcal{A}_m(\mathbf{v})$.

In case (iii) the shifted vectors are nonnegative automatically. A shifted vector is forbidden precisely when it lies below $\mathbf{v}$ and is not equal to β_* . Apart from β_* , no admissible target with at least three coordinates in $\text{Hi}(\beta)$ is mandatory.

Proof. The center has a unique special target, proving case (i). Every one-piece output of an admissible target has the form $\mathbf{c} = \beta + m\mathbf{e}_j$ and is different from $\mathbf{v}$, since otherwise the center would have an admissible one-piece representation. Its eligible coordinates are exactly $\{j\} \cup \text{Hi}(\beta)$.

If $\text{Hi}(\beta)$ is empty, any choice of j gives β as the unique predecessor. If it has one member, choose that member as j . This proves case (ii).

If $\text{Hi}(\beta) = \{p, q\}$, taking $j = p$ gives exactly two predecessors, β and $\beta + m\mathbf{e}_p - m\mathbf{e}_q$. Thus β is the unique admissible predecessor precisely when the second is forbidden. Taking $j = q$ gives the other transfer. Any different j creates at least three eligible coordinates. By Lemma 29.2, at most one predecessor is below the center and therefore at most one is forbidden. So at least two are admissible, and no singleton constraint arises. This proves (iii).

Finally, if $|\text{Hi}(\beta)| \geq 3$, every one-piece output of β has at least three predecessors and again at least two are admissible. Unless β is the unique special target, Proposition 33.1 makes it optional. The description of forbidden shifts follows from the unique special target and (54). $\square$

This supplies an explicit answer to the mandatory-word part of Question 21.4. It does not equate the mandatory set with an entire realizing language. Additional choices may be necessary to hit predecessor sets of size two or more.

Table 4: Maximal and mandatory target counts for the small extremal centers. The maximal counts are not asserted to be minima.

m	center type	$ \mathcal{A}_m(\mathbf{v}) $	mandatory targets
3	(4, 4, 4)	46	28
4	(8, 8, 1, 1)	665	263
4	(6, 6, 5, 1)	656	257
5	(10, 10, 1, 1, 1)	7 280	3 080
5	(7, 7, 7, 1, 1)	7 255	3 066

These finite counts were obtained from the explicit theorem and checked independently by enumerating every admissible-predecessor set. The next section solves the remaining choices completely in the first row.

34 Complete classification at degree three

At the shortest length for $m = 3$, the center is uniquely $\mathbf{v} = (4, 4, 4)$, the total is 12, and target gap vectors have total 9. The largest target set is

$$\mathcal{A} = \{\beta \in C_{9,3} : \max_i \beta_i \geq 5\} \cup \{(3, 3, 3)\}. \quad (80)$$

It has size 46, agreeing with Part II's full construction: it is the target set of the worked example (39) (Theorem 17.1). Here we determine exactly which subsets realize the center.

For a tuple u , write $\text{Perm}(u)$ for the set of its distinct coordinate permutations. Define

$$\begin{aligned} \mathcal{M} = \{(3, 3, 3)\} \cup \text{Perm}(0, 0, 9) \cup \text{Perm}(0, 1, 8) \cup \text{Perm}(0, 2, 7) \\ \cup \text{Perm}(1, 1, 7) \cup \text{Perm}(1, 2, 6) \cup \text{Perm}(2, 2, 5). \end{aligned} \quad (81)$$

The six permutation sets in the second part have respective sizes 3, 6, 6, 3, 6, 3, so $|\mathcal{M}| = 28$.

The remaining 18 targets divide into nine pairs. Three pairs come from outputs in $\text{Perm}(0, 6, 6)$: the two nonzero coordinates allow the choices $(0, 3, 6)$ and $(0, 6, 3)$ in their respective positions. Six pairs come from outputs $\mathbf{c} \in \text{Perm}(3, 4, 5)$: delete three from the coordinate equal to three or from the coordinate equal to four. For example, the output $(3, 4, 5)$ gives the pair

$$\{(0, 4, 5), (3, 1, 5)\}. \quad (82)$$

Let $\mathcal{P}_1, \dots, \mathcal{P}_9$ denote these nine two-element sets.

Theorem 34.1 (Every shortest degree-three language). *The nine pairs are pairwise disjoint and partition $\mathcal{A} \setminus \mathcal{M}$. A target set $\mathcal{E} \subseteq C_{9,3}$ realizes the unique center $(4, 4, 4)$ if and only if*

$$\mathcal{E} = \mathcal{M} \cup \mathcal{E}_1 \cup \dots \cup \mathcal{E}_9, \quad \emptyset \neq \mathcal{E}_i \subseteq \mathcal{P}_i. \quad (83)$$

Consequently there are exactly $3^9 = 19\,683$ shortest-length target languages. Their size-generating polynomial is

$$\sum_{\mathcal{E} \text{ realizing}} z^{|\mathcal{E}|} = z^{28} (2z + z^2)^9 = z^{37} (2 + z)^9. \quad (84)$$

The smallest cardinality is 37, attained by exactly $2^9 = 512$ languages. More generally, for $0 \leq j \leq 9$, exactly $\binom{9}{j} 2^{9-j}$ languages have size $37 + j$.

Proof. The nondecreasing target types in (80) are

$$\begin{aligned} (0, 0, 9), (0, 1, 8), (0, 2, 7), (0, 3, 6), (0, 4, 5), (1, 1, 7), \\ (1, 2, 6), (1, 3, 5), (2, 2, 5), (3, 3, 3). \end{aligned}$$

To see that the list is complete, put $x \leq y \leq z$, $x + y + z = 9$, and $z \geq 5$. If $x = 0$, then $0 \leq y \leq 4$; if $x = 1$, then $1 \leq y \leq 3$; if $x = 2$, only $y = 2$ is possible; and $x \geq 3$ is impossible with $z \geq 5$. Add the exceptional target $(3, 3, 3)$ separately.

Every target in $\mathcal{M}$ is mandatory. The special target is forced by the center; each other listed target has at most one coordinate at least three, so Theorem 33.2 applies. The other targets are precisely

$$\text{Perm}(0, 3, 6) \cup \text{Perm}(0, 4, 5) \cup \text{Perm}(1, 3, 5),$$

each of size six. The first six split into the three pairs associated to $\text{Perm}(0, 6, 6)$. Each target of type $(0, 4, 5)$ has a unique paired target of type $(1, 3, 5)$: increase the zero to three and decrease the coordinate four to one. This is a bijection, producing the other six pairs. Hence all nine pairs are disjoint and cover the optional targets.

For an output of type $(0, 6, 6)$, the two displayed predecessors are its only one-piece predecessors, and both are admissible. For type $(3, 4, 5)$, the third predecessor, obtained by deleting from the five, has type $(2, 3, 4)$. It lies below the center and is not $(3, 3, 3)$, so it is forbidden. Thus in each of these nine outputs a realizing language must choose a member of the associated pair.

It remains to prove that these constraints suffice for all other outputs. Table 5 lists every nondecreasing output triple of total 12. For each type other than $(0, 6, 6)$, $(3, 4, 5)$, and the center, it gives a predecessor in $\mathcal{M}$. Coordinate permutation preserves both the mandatory set and the predecessor relation, so the table covers every ordered output. The list is exhaustive by taking its smallest coordinate successively equal to $0, 1, 2, 3, 4$. The center has its mandatory special target. Therefore Corollary 27.3 proves sufficiency.

There are three nonempty subsets of each pair, independently, so the number of languages is 3^9 . Each pair contributes $2z + z^2$ to its cardinality polynomial, while the mandatory set contributes z^{28} . Expanding gives (84) and the remaining assertions. $\square$

Table 5: Exhaustive degree-three output audit up to coordinate permutation. An ordinary listed predecessor belongs to $\mathcal{M}$.

Output	mandatory predecessor or remaining constraint	Output	predecessor
$(0, 0, 12)$	$(0, 0, 9)$	$(1, 3, 8)$	$(1, 0, 8)$
$(0, 1, 11)$	$(0, 1, 8)$	$(1, 4, 7)$	$(1, 1, 7)$
$(0, 2, 10)$	$(0, 2, 7)$	$(1, 5, 6)$	$(1, 2, 6)$
$(0, 3, 9)$	$(0, 0, 9)$	$(2, 2, 8)$	$(2, 2, 5)$
$(0, 4, 8)$	$(0, 1, 8)$	$(2, 3, 7)$	$(2, 0, 7)$
$(0, 5, 7)$	$(0, 2, 7)$	$(2, 4, 6)$	$(2, 1, 6)$
$(0, 6, 6)$	choose $(0, 3, 6)$ or $(0, 6, 3)$	$(2, 5, 5)$	$(2, 2, 5)$
$(1, 1, 10)$	$(1, 1, 7)$	$(3, 3, 6)$	$(3, 3, 3)$
$(1, 2, 9)$	$(1, 2, 6)$	$(3, 4, 5)$	pair (82)
		$(4, 4, 4)$	center; degree three

34.1 An explicit 37-target realization

No large word listing is needed to specify a minimum language. Take the 28 targets in (81) and the lexicographically smaller gap vector from each of the nine pairs. The selected additional vectors are

$$\begin{aligned} &(0, 3, 6), (0, 4, 5), (0, 5, 4), (1, 3, 5), (1, 5, 3), \\ &(3, 0, 6), (3, 6, 0), (5, 0, 4), (5, 1, 3). \end{aligned} \tag{85}$$

Together these specify $\mathcal{E}$ and hence the binary language $B_{\mathcal{E}}$ via (48). Its shuffle has 91 distinct outputs. Ninety have degree one, and the sole degree-three output is $a^4ba^4ba^4$. All 37 actual target words are listed in `data/03-sharp-isolation-minimal-degree3-targets.csv`. Independent position-mask enumeration checked this example without using the gap-degree formula: 13 468 masks produce exactly the stated 91 words and degrees.

34.2 What the cardinality result does and does not optimize

The minimum 37 is over all target languages at the globally shortest output length for $m = 3$. This is stronger than optimizing a chosen safe construction, because Theorem 31.1 first forces the unique center and Lemma 27.2 includes every possible realizing target language. It is not a theorem that 37 is the minimum target count if longer output lengths or incomplete fixed-letter-count coverage are permitted. Those are different optimization problems.

35 Residue classes and the remaining covering problem

The language characterization (56) gives a finite optimization problem once a center is fixed. There is additional structure: one-piece insertion adds m to one coordinate, so all one-piece constraints preserve the entire coordinatewise residue vector modulo m .

Proposition 35.1 (Residue decomposition). *Fix a feasible center $\mathbf{v} \in C_{T,g}$. For $\mathbf{r} \in \{0, \dots, m-1\}^g$ with*

$$\mathbf{v} = \frac{T - \sum_i r_i}{m} \in \mathbb{N}_0,$$

outputs in residue class $\mathbf{r}$ are exactly $\mathbf{c} = \mathbf{r} + m\mathbf{z}$ with $\mathbf{z} \in C_{\mathbf{v},g}$. Targets in that residue class are $\boldsymbol{\beta} = \mathbf{r} + m\mathbf{y}$ with $\mathbf{y} \in C_{\mathbf{v}-1,g}$. Their one-piece relation is

$$\mathbf{y} \mapsto \{\mathbf{y} + \mathbf{e}_j : 1 \leq j \leq g\}. \quad (86)$$

After deleting forbidden targets and omitting the exceptional output from the degree-one requirements, the covering constraints split into independent problems for the residue classes. When $g = m$, the sole special target is forced in its class and minimum target cardinalities add over classes.

Proof. Each nonnegative coordinate has a unique residue in $\{0, \dots, m-1\}$. Taking sums proves the displayed total constraints. Adding $m\mathbf{e}_j$ to a target adds $\mathbf{e}_j$ to its quotient vector and leaves its residues unchanged. Thus no predecessor constraint can involve targets in two residue classes. When there is one special target, its required inclusion is a local constraint in just one class, so both feasible choices and the cardinality objective separate across classes. $\square$

If $g > m$ and several special targets exist, the ordinary one-piece constraints still separate. The requirement that at least one special target be accepted can couple the choices of residue classes, so the final additive conclusion must not be applied unchanged in that case.

At the shortest lengths with $m \geq 3$, one has $\nu \leq m-1$ for every nonempty residue class when $m \geq 6$, since $T_m = m^2 - m + 1$ and $\nu \leq \lfloor T_m/m \rfloor = m-1$. The analogous bounds are $\nu \leq 4$ for $m = 3, 4, 5$. Thus the decomposition replaces one large fixed-count class by many small simplex-layer incidence problems. The bounds do not imply a polynomial-time algorithm in m : the number of residue classes and quotient vectors may still be large.

Corollary 35.2 (A general counting lower bound for targets). *Every realizing target set satisfies*

$$|\mathcal{E}| \geq \max \left\{ \left\lceil \frac{\binom{T+g-1}{g-1} - 1}{g} \right\rceil, \text{number of mandatory targets} \right\}. \quad (87)$$

Proof. There are $\binom{T+g-1}{g-1} - 1$ nonexceptional outputs, each requiring a one-piece representation. Each target has exactly g distinct one-piece outputs, obtained by adding m in one of its coordinates. The union bound gives the first inequality; mandatory inclusion gives the second. $\square$

For the degree-three shortest problem, these elementary bounds give only 30 and 28, whereas the disjoint-pair obstruction raises the exact answer to 37. This illustrates the importance of overlap constraints.

36 Reproducible exact checks

36.1 What was run

The manuscript was delivered with a standard-library Python implementation, shipped here as `code/03-sharp-isolation-isolation.py` and `code/03-sharp-isolation-verify.py`. Its universal claims rest on the proofs above, not on numerical extrapolation. The finite audits deliberately compare different representations of the same objects. The execution completed with status `PASS`; exact ranges and counts are recorded in `data/03-sharp-isolation-verification.json` (delivered as `data/verification.json`).

Table 6: Executed verification suites. Samples and exhaustive ranges are kept distinct.

Suite	Scope and recorded outcome
Literal source masks	All 508 pairs with $m = 1, 2, 3, 4$ and binary target length 0–6; 30 464 masks and 8 344 distinct-word queries agree with the gap formula.
Complete center-orbit range	All 1 907 nondecreasing center vectors with $m = 3, 4, 5$, $g = 1, 2, 3, 4$, and $m \leq T \leq 16$; direct output search agrees with the classification. There are 69 343 inspected outputs, with failed cases stopping at their first counterexample.
Independent deficit search	588 centers with $m = 3$ –6 and $g = m, m + 1, m + 2$: 40 seeded random and 9 prescribed centers per parameter pair; 191 114 deficit queries agree with the scalar criterion.
Complete explicit examples	Every output of all five small extremal types and the boundary centers for $m = 3, 4, 5$; the largest checked example has 324 632 outputs.
Center counting	Formula versus independent permutation-orbit enumeration for $m = 3$ –7; bounded-composition inclusion–exclusion versus direct coordinate enumeration; formula values recorded through $m = 12$.
Mandatory targets	Explicit transfer test versus singleton-predecessor constraints for every target of the five small extremal types.
Degree-three languages	All $3^9 = 19\,683$ matching choices checked; one 37-target example also checked by 13 468 literal masks.
Parameter construction	Every $m = 3$ –60, $g = m$ through $m + 30$, and total increments 0, 1, m , m^2 above the threshold: 7 192 constructed triples.

The center-orbit enumeration is sufficient for its stated finite range because all tests are invariant under coordinate permutation. It is not an enumeration of every target-language subset. The larger-dimension deficit suite uses a fixed random seed, 20260930, and is explicitly a sample. The parameter-construction suite checks the formulas at finitely many inputs, while Theorem 30.2 proves them for all inputs.

36.2 Files and execution

The manuscript was delivered with its standalone article, its compiled PDF, source and proof audits, and the following computational files, shipped in this report under the prefixed names given here (the delivered name is the part after the prefix; Section 40 maps every file):

`code/03-sharp-isolation-isolation.py`: definitions, classifier, maximal target generator, independent output and deficit checks, counting formulas, mandatory-target test, and literal mask enumeration.
`code/03-sharp-isolation-verify.py`: complete audit driver and data exporter.
`data/03-sharp-isolation-extremal_center_counts.csv`: the exact count table.

`data/03-sharp-isolation-extremal_orbits_m3_m6.json`: all extremal orbits through degree six.
`data/03-sharp-isolation-complete_examples.csv`: full finite-coverage example receipts.
`data/03-sharp-isolation-degree3_pair_constraints.json`: the nine explicit pairs.
`data/03-sharp-isolation-minimal_degree3_targets.csv`: a minimum 37-word target language.
`data/03-sharp-isolation-verification.json`: ranges, counts, and executed status.
`data/03-sharp-isolation-verification_run.txt`: the corresponding console output.

The manuscript’s instruction for rerunning without overwriting the shipped receipt was to pass `-output-dir` to its verifier. In this report that is not enough on its own: the shipped verifier imports its companion module under the delivered name `isolation`, and its default output directory is this report’s `data/`, where it would overwrite Part I’s `data/verification.json`. Section 40.2 gives a recipe that runs on a copy. No theorem prover, numerical optimizer, SMT solver, external package, or network connection is needed for the checks. Python 3.10 or later is required. The code uses exact integers; the sampled suite is reproducible. The delivered Makefile, shipped as `code/03-sharp-isolation-Makefile`, compiles the delivered article with pdfLaTeX and runs the verification into a separate local directory; it uses the delivered layout and is not meant to be run from this report (Section 40.2).

37 Proof audit and formalization boundary

37.1 Dependencies

The proof structure is short enough to audit without trusting a computer algebra system. The gap formula gives the maximal admissible target set. Support and capped compositions give the volume obstruction. The unique-predecessor deficit analysis gives the complete center criterion. An integer optimization over the number of high coordinates gives the mass bound, and explicit padded centers attain every numerical threshold. Mandatory targets come from singleton covering constraints and the fact that at most one predecessor can be unsafe. The degree-three language classification then uses a complete list of 19 unordered output types.

No result about real-rooted polynomials, semilinear sets, decidability of context-free languages, or proof-assistant soundness enters these arguments. The cited preprint and Parts I–II fix the problem and the comparison point; every needed mathematical ingredient is proved here or, where printed once, in Part II.

37.2 Potential failure modes checked explicitly

The following distinctions are essential.

Degree is minimized over targets and masks. A target word may admit many insertion masks. The gap lemma proves the minimum for that target, and only then minimizes across the language. No uniqueness of a mask is assumed.

Full coverage is used in both lower bounds. A short-gap word with no eligible coordinate must still be an output. That is the source of the volume obstruction. Without coverage, omitting such words can evade the argument.

The total inequality is strict. The capped boundary is excluded by a concrete neighbor of support two. A large excess-height sum alone cannot replace the volume test.

The strict escape criterion is not necessary by itself. The balanced boundary family is a counterexample. The classification retains the exact squarefree-deficit alternative rather than silently discarding it.

The length minimum is over all numbers of gaps. The parameter theorem and the strictly increasing fixed-gap length formula rule out smaller witnesses with extra gaps. The small-case proof is not a search restricted to $g = m$.

Counts refer to different objects. The values e_m count ordered center vectors. The values 3^9 and 2^9 count distinct target languages for the sole shortest degree-three center. Neither is a count of source masks or an identification modulo word reversal.

Maximal does not mean minimum. The canonical language $\mathcal{A}_m(\mathbf{v})$ is the largest admissible one. Its cardinality is a useful audit, not an optimality claim. Only the degree-three shortest case has its target minimum solved here.

37.3 A realistic formalization route

The mathematical statements are finite and combinatorial. A prospective formalization can use functions from a finite index type to natural numbers, finite sums, coordinate comparison, and support cardinality. The main interfaces would be the gap-word encoding, a literal insertion-mask relation, and the theorem identifying minimum runs with support. The remaining classification can then be stated entirely on finite vectors.

A small independently checked development could first formalize the capacity-completion lemma used in Lemma 29.4: a bounded nonnegative vector can be raised to any intermediate total up to the sum of its capacities. It could next formalize the maximal-language reduction and the local obstruction equivalence. The arithmetic minimization in Lemma 30.1 and the degree-three table are separate, bounded obligations. A reflected checker for explicit centers would be useful, but proving its soundness must precede treating its output as a proof certificate.

No Lean or Rocq code is included as though it had been checked. No theorem in this manuscript inherits formal status from unrelated developments in ProveIt. The accompanying Python audits are executable checks, not proof-assistant verification.

38 Further research questions and topics

The questions below distinguish remaining problems of Part II from new problems suggested by this classification. Unless a source is named, they are proposed directions, not claims that an exhaustive literature survey has established their global open status.

Question 38.1 (Minimum target size at higher degrees). For each extremal center with $m \geq 4$, determine the minimum number of targets and the number of minimum languages. The explicit mandatory-target test and residue decomposition reduce this to a concrete finite covering problem. Already the two center types at $m = 4$ have different mandatory counts, so minimizing center length does not identify the best target size. A useful first goal is a rigorous certificate for both $m = 4$ types. This continues Question 21.2.

Question 38.2 (Trading length against target count). For fixed m , determine the Pareto frontier of output length and target cardinality under full coverage and one exception. In particular, can a longer degree-three construction use fewer than 37 targets? The theorem here fixes the shortest length; it does not exclude such a tradeoff.

Question 38.3 (Asymptotic density of extremal centers). Determine the asymptotic growth of e_m and its ratio to $\binom{T_m-1}{m-1}$, the number of all positive center vectors of total T_m . The finite sum (77) gives an exact starting point. A probabilistic analysis of the excess mass outside the largest coordinate may expose whether feasible centers are typical or rare at the critical total.

Question 38.4 (Several exceptional words without separation). Classify two or more exceptional vectors whose forbidden-predecessor neighborhoods overlap. Part II's separated-center construction (Theorem 14.4) is sufficient, but for several centers the fact that at most one predecessor lies below a *fixed* center no longer bounds the total number forbidden by all centers. A sharp two-center criterion would be the next structural step. This continues Question 21.5.

Question 38.5 (An isolated degree below the inserted length). Keep the inserted word a^m , but require the unique exception to have degree r with $2 \leq r < m$. Admissible deficits then have support at least r , and at least one accepted deficit must have support exactly r . The binary simplicity of a total- m support- m deficit is lost. Determine the analogue of the height classification and the exact threshold in (m, r, g, T) .

Question 38.6 (Removing full coverage; restates Question 21.6). Determine the shortest binary realizations of spectrum $\{1, m\}$ with $A = \{a^m\}$ when the shuffle may omit words of the fixed-count class. This is Part II's Question 21.6. The quadratic lower bound in the full-coverage problem does not apply. A successful construction would have to omit low-eligibility outputs while controlling all outputs actually created by its targets.

Question 38.7 (Mandatory targets versus sufficient cores). Which centers have a mandatory target set that already realizes the center? When it does not, can the remaining predecessor hypergraph be characterized as a matching, a graph of bounded width, or another tractable family? The nine-edge matching at degree three suggests seeking structural explanations for exact target-count polynomials rather than relying only on optimization.

Question 38.8 (Beyond one unary inserted word; restates Question 21.1). The unrestricted both-singleton insertion-degree conjecture is Hughes's Conjecture 2 [1, Section 11] and Part II's Question 21.1. A first extension beyond the present gap model is an inserted word $a^p b^q$. Its b 's no longer all come from the target, so the skeleton is not fixed. Identify a replacement for the gap minimum that retains enough structure to address that case. No conclusion about this conjecture follows from the present classification.

Question 38.9 (Weighted and constrained gaps). Assign a positive cost to each receiving gap, or allow insertion only in a specified family of gaps. The target degree becomes a weighted support minimum. Determine which weight systems admit a maximal-language reduction with an equally simple obstruction test, and which turn the local completion problem into a genuinely difficult optimization. This overlaps the weighted extension in Question 21.10.

Question 38.10 (A formally checked classification theorem). Formalize the complete center criterion and the exact parameter threshold, then connect the vector model to literal word insertion. The explicit negative certificates and the complete degree-three table offer small independent checking targets. An end-to-end result should establish the language semantics, not merely verify the numerical height inequality. It overlaps the Lean programme sketched in Question 21.10.

39 Conclusion of Part III

Part II's binary-profile construction becomes an exact extremal theory once the largest admissible target language is identified and unique-predecessor outputs are analyzed without a strictness shortcut. This yields all feasible parameter triples, proves optimality of lengths 14, 21, 27, classifies every sharp-bound center, and supplies an explicit mandatory-target test. At degree three the residual constraints are a disjoint matching, leading to the exact minimum 37 and the full language-count polynomial.

These are concrete answers to Part II's small-witness and classification questions (Questions 21.3 and 21.4). Their scope is deliberately specific. The remaining target-covering problems, multiple exceptions, non-unary sources, and loss of full coverage require additional arguments rather than an extrapolation of the present theorem.

40 Package contents, rebuilding and the choices of the merge

40.1 Shipped files

Manuscript 04 was delivered with its source `article.tex`, its compiled PDF, a delivery README, `PROOF_AUDIT.md`, `SOURCE_AUDIT.md`, a Makefile, two Python files in `code/` and nine data files in `data/`; there was no checksum file. The report ships, beside the files of Parts I and II:

Shipped path	Delivered as
<code>code/03-sharp-isolation-isolation.py</code>	<code>code/isolation.py</code>
<code>code/03-sharp-isolation-verify.py</code>	<code>code/verify.py</code>
<code>code/03-sharp-isolation-Makefile</code>	Makefile
<code>data/03-sharp-isolation-verification.json</code>	<code>data/verification.json</code>
<code>data/03-sharp-isolation-verification_run.txt</code>	<code>data/verification_run.txt</code>
<code>data/03-sharp-isolation-extremal_center_counts.csv</code>	<code>data/extremal_center_counts.csv</code>
<code>data/03-sharp-isolation-extremal_orbits_m3_m6.json</code>	<code>data/extremal_orbits_m3_m6.json</code>
<code>data/03-sharp-isolation-complete_examples.csv</code>	<code>data/complete_examples.csv</code>
<code>data/03-sharp-isolation-degree3_pair_constraints.json</code>	<code>data/degree3_pair_constraints.json</code>
<code>data/03-sharp-isolation-minimal_degree3_targets.csv</code>	<code>data/minimal_degree3_targets.csv</code>
<code>data/03-sharp-isolation-provenance.json</code>	<code>data/provenance.json</code>
<code>data/03-sharp-isolation-build_status.json</code>	<code>data/build_status.json</code>
<code>03-sharp-isolation-PROOF_AUDIT.md</code>	<code>PROOF_AUDIT.md</code>
<code>03-sharp-isolation-SOURCE_AUDIT.md</code>	<code>SOURCE_AUDIT.md</code>

The delivered article, PDF and README are printed or replaced by this report and its README. The shipped files are byte-identical to the delivered ones; the three CSV files keep their delivered CRLF line endings.

40.2 Rebuilding and rerunning

The article is rebuilt from the report directory with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Part III verifier needs Python 3.10 or later and only its standard library. *It cannot be run in place.* It imports its companion module under the delivered name `isolation`, which the prefixed file does not provide, and its default output directory is the `data/` directory next to its own `code/` directory: in this report that would overwrite Part I's `data/verification.json` and add unprefixed files. Run it on a copy with the delivered names and an explicit output directory, for example

```
mkdir -p <scratch>/code
cp code/03-sharp-isolation-isolation.py <scratch>/code/isolation.py
cp code/03-sharp-isolation-verify.py <scratch>/code/verify.py
cd <scratch>
python code/verify.py --output-dir audit-output
```

and compare `audit-output/` with the shipped `data/03-sharp-isolation-*` files. The verifier rewrites `verification.json`, the two other JSON files and the three CSV files; it prints, but does not write, the console record, and it does not regenerate `provenance.json` or `build_status.json`. On Windows the JSON files are written with CRLF line endings.

[Added in the merge: the suite was run this way on a copy on 30 September 2026 with Python 3.14.4, during the batch-66 placement checks. It passed in about seven seconds; its three CSV files were byte-identical to the shipped ones, its two other JSON files equal apart from line endings, and its `verification.json` equal to the shipped one apart from line endings and the field `python` (3.14.4 instead of 3.13.5).]

The shipped `code/03-sharp-isolation-Makefile` still uses the delivered layout: its `verify` target runs `code/verify.py`, which in this report is Part I’s verifier, and its `pdf` target runs pdfLaTeX on the merged `article.tex` into `build/`. It is shipped as delivered and should not be used from the report directory.

40.3 Where the merge had to choose

1. Placement: the manuscript answers questions of Part II, so it became Part III of this report rather than a separate report. It is printed after Part II and before Part I’s appendices, which contain no numbered items, so no number of Part I or Part II moved.
2. What the manuscript repeats from Parts I and II is printed once (Section 25.8). The gap-degree lemma keeps its statement and points to Part II for the proof; the volume proposition keeps the manuscript’s proof, marked as such, because its route to $g \geq m$ differs.
3. Seven symbols were renamed where they clash with Parts I and II, and the manuscript’s $C(T, g)$, $\mathcal{W}(T, g)$ and $I_k(A, B)$ are printed with the report’s $C_{T,g}$, $\mathcal{W}_{T,g}$ and $\mathcal{I}_k(A, B)$ (Table 2); no normalization changed.
4. The manuscript’s title page, abstract, two boxed statements and Section 1 became Sections 25.1–25.6; the boxes are printed as plain text. Its title-page wording naming the assistant and the addressee is recorded in Section 25.1 only.
5. The manuscript’s references to the pinned report by number (Theorem 12.3, Section 16, Questions 21.1, 21.3, 21.4, 21.6) became internal references, and its ten unlabelled questions received labels and pointers to the Part II questions they restate or continue (Questions 38.1–38.10).
6. Its file names and rerun instructions were replaced by the shipped names and the copy-based recipe of Section 40.2.
7. The introductory note of Part III, Sections 25.7 and 25.8, this section, and the dated pointers in Parts I and II were added in the merge.

A The binary finite certificate

For each source pair in (10), the table records the number of *distinct* output words whose singleton minimum has the indicated value. It is obtained by enumerating all thirty-five masks, not by counting masks as distinct outputs. Every row has nonzero counts at every degree from 1 through its maximum.

u	v	$d = 1$	$d = 2$	$d = 3$	$d = 4$
aaba	aaa	4	0	0	0
aaba	aba	4	2	0	0
aaba	bab	4	8	1	0
aabb	aaa	4	6	0	0
aabb	aba	4	6	0	0
aabb	bab	4	6	0	0
abbb	aaa	4	12	4	0
abbb	aba	4	7	0	0
abbb	bab	4	4	0	0
bbaa	aaa	4	6	0	0
bbaa	aba	4	6	0	0
bbaa	bab	4	6	0	0
bbba	aaa	4	12	4	0
bbba	aba	4	7	0	0
bbba	bab	4	4	0	0
bbbb	aaa	4	18	12	1
bbbb	aba	3	9	3	0
bbbb	bab	2	3	0	0

Across these rows, the union of the singleton degree-three words contains twenty-three words. For each, the next table provides a competing source pair and a mask of cost at most two. Reading positions marked A or B reconstructs the displayed sources, so each row is directly checkable.

Output w	$u \in A$	$v \in B$	Mask	Degree
aababab	aaba	bab	AAAABBB	1
abaabab	aaba	bab	ABBAAAB	2
abaabba	aabb	aba	BBAAAAB	1
ababaab	aaba	bab	ABAAABB	2
abababa	aaba	bab	ABBBAAA	2
abababb	aabb	bab	ABBBAAA	2
ababbab	aabb	bab	ABAAABB	2
abbabab	abbb	aba	AAABBBB	2
baababa	aaba	bab	BAAABBA	2
baababb	aabb	bab	BAAABAB	2
baabbab	aabb	bab	BAAAABB	1
babaaba	aaba	bab	BBBAAAA	1
babaabb	aabb	bab	BBBAAAA	1
bababaa	bbaa	aba	ABBBAAA	2
bababba	bbba	aba	ABBBAAA	2
babbaab	bbaa	bab	BBAAAAB	1
babbaba	bbaa	bab	BBAAABA	2
babbabb	abbb	bab	BAAABAB	2
babbbab	abbb	bab	BAAAABB	1
bbaabab	bbaa	bab	AAAABBB	1
bbabaab	bbaa	bab	AAABABB	2
bbababa	bbaa	bab	AAABBBB	2
bbabbab	bbba	bab	ABBAAAB	2

These tables certify the missing degree as follows. A language-level degree-three output would have a

singleton representation of minimum degree exactly three for some pair. It would therefore occur in the second table, which supplies a lower-cost representation, a contradiction. Degree four is witnessed by *bababab* as proved in Section 7. The complete degree map, including all fifty-seven degree-one and fifty-three degree-two outputs, is in `data/binary_degrees.csv`.

B Archive contents and rebuilding

The archive contains the complete $\text{\LaTeX}$ source, the compiled PDF, the included table sources, the exact Python verifier, a portable search-instance generator, an SMT-LIB file, the verified degree tables, and a machine-readable verification report. The small examples and all input constants are embedded in the verifier, so no external data downloads are necessary. [Added 29 September 2026, batch 41: this appendix describes Part I’s files. Part II’s files carry the prefix `02-binary-profiles-`; Section 24 lists them and explains how to rerun their verifier without overwriting Part I’s `data/verification.json`.] [Added 30 September 2026, batch 66: Part III’s files carry the prefix `03-sharp-isolation-`; Section 40 lists them and explains how to rerun their verifier on a copy.]

From the archive’s root directory, verification is:

```
python code/verify.py
```

This rewrites the data tables and the JSON report. Rebuilding the article uses a conventional TeX Live or MiKTeX installation with the packages named in the preamble:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is embedded, so BibTeX is not required. The generated table fragments are included in the archive. The verifier needs Python 3.9 or newer and no third-party package. Optional discovery experiments use an external SMT solver; all mathematical results and verification commands remain available without it.

References

- [1] Charles E. Hughes. *Undecidability of Adjacent Equality for Insertion, Shuffle, and Crossover Language Operations*. arXiv:2608.27755v1, 27 August 2026. <https://arxiv.org/abs/2608.27755v1>. Sections 10–11 contain the no-gap questions addressed here.
- [2] A. Mateescu, Grzegorz Rozenberg, and Arto Salomaa. Shuffle on trajectories: Syntactic constraints. *Theoretical Computer Science* **197**(1–2), 1–56, 1998. [https://doi.org/10.1016/S0304-3975\(97\)00163-1](https://doi.org/10.1016/S0304-3975(97)00163-1).